

INTEGRABLE METRICS AND EFFECTIVITY WITH CONTROLLED BOUNDARY

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ABSTRACT. Let Y be smooth projective and let C be an effective integral divisor. If $-K_Y + C$ admits a metric with a global strict curvature lower bound for which the canonical section of C is locally square integrable, every pseudoeffective rational divisor becomes rationally effective after an effective correction supported on C . For a smooth projective X with smoothly semipositive $L = -K_X$, sections of $m_j L - P$ at unbounded positive exponents, with any fixed pseudoeffective Cartier error P , yield a section of a positive multiple of L .

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Date: September 26, 2026.

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1. INTRODUCTION

A pseudoeffective divisor need not have a section in any positive multiple. This paper studies an analytic condition that makes such a divisor effective up to rational linear equivalence, after a correction on a prescribed boundary. The condition is integrability of the canonical divisor section for a positively curved anticanonical metric. Its usefulness is birational: a correction on the base of a fibration can disappear after pullback and pushforward to the original variety.

If the squared norm in a local frame of a singular metric h is $e^{-\phi}$, its multiplier ideal $\mathcal{J}(h)$ consists of the holomorphic germs u with $|u|^2 e^{-\phi}$ locally integrable. Thus $\mathcal{J}(h) \supset \mathcal{O}_Y(-C)$ means precisely that the canonical section s_C of the effective integral divisor C has locally finite squared norm. We call a rational divisor *rationally effective* if it is rationally linearly equivalent to an effective rational divisor. A rational divisor is *pseudoeffective* when its numerical class lies in the closure of the cone generated by effective divisors. On a smooth projective variety this is equivalent to the existence of a singular Hermitian metric with semipositive curvature current on its rational line bundle.

Theorem 1.1 (Effectivity modulo a controlled boundary). *Let Y be a smooth projective complex variety of positive dimension and let $C \geq 0$ be an integral divisor. Suppose $-K_Y + C$ has a singular Hermitian metric h whose curvature dominates a Kähler form and for which*

$$\mathcal{J}(h) \supset \mathcal{O}_Y(-C).$$

For every pseudoeffective rational divisor M on Y , there is an effective rational divisor C_0 supported on C such that $M + C_0$ is rationally effective.

The support conclusion distinguishes the theorem from numerical nonvanishing. It permits the correction to be removed under a specified birational map. The analytic assumption also distinguishes two notions of integrability: s_C may have finite norm even when a nonvanishing local frame does not. The stronger, unweighted condition will yield a separate finite-generation result.

1.1. The boundary construction and its antecedents. An effective rational boundary Δ is *klt* (Kawamata log terminal) if, on a log resolution $\sigma : Y' \rightarrow Y$, every coefficient of $\sigma^*(K_Y + \Delta) - K_{Y'}$ is less than one. For a simple normal crossing boundary on a smooth variety, this amounts to each boundary coefficient being less than one.

The proof connects multiplier-ideal approximation with big-boundary nonvanishing. Nadel vanishing [25], followed by very ample regularity, makes the relevant multiplier-ideal systems globally generated. Point extension of Ohsawa–Takegoshi [26], in the weighted Bergman approximation developed by Demailly [15], compares the fixed divisor of a complete system with the metric’s

singularities. On a resolution $\sigma : Y' \rightarrow Y$, weighted integrability produces a signed divisor $\Psi = \Psi^+ - \Psi^-$ whose coefficients are strictly below one. Its negative part lies only over C and the exceptional locus.

The approximation leaves an ample rational class A in reserve. For a prescribed pseudoeffective M , choose a small rational $t > 0$ so that $A + tM$ remains ample. General free and ample representatives then turn the positive part of the signed divisor into an *effective big klt* boundary Δ , with

$$K_{Y'} + \Delta \sim_{\mathbb{Q}} t\sigma^*M + \Psi^-.$$

This is the pair to which the nonvanishing theorem of Birkar–Cascini–Hacon–McKernan [8, Theorem D] applies. A finite rational linear system upgrades its real-linear conclusion to rational linear equivalence. Pushforward gives the correction $\sigma_*\Psi^-/t$, supported on C . Sections 2 and 3 give this construction. The latter also retains an analytic perturbation argument: strong openness of Guan–Zhou [19] and small-exponent integrability of Skoda [33] permit a small pseudoeffective perturbation of the metric while preserving the integrability of the distinguished section.

A different consequence begins with an unweighted integrable semipositive metric that is smooth and strictly positive on one ordinary open set. Boucksom’s volume criterion [9] first gives bigness; a small mixture with a strict metric then gives a global curvature lower bound while preserving integrability. Section 4 makes this preparation explicit and permits perturbation in finitely many arbitrary rational directions. The resulting klt boundaries fit the multigraded adjoint finite-generation results of Cascini–Lazić [13] and Corti–Lazić [14, Theorem 3.2(1)]. More precisely, if $-K_Y + A_0$ has this unweighted metric, then for every finite list of rational divisors $P_1, \dots, P_s$ there are a common rational $\epsilon > 0$ and Cartier multiples $B_i = k_0(A_0 + \epsilon P_i)$ whose multigraded section ring is finitely generated (Theorem 4.3). This controls the sections of specified Cartier divisors, not merely a numerical cone.

1.2. Anticanonical nonvanishing and fixed-error conversion. For a smooth projective variety X , anticanonical nonvanishing asks whether some $H^0(X, -mK_X)$, $m > 0$, is nonzero when $-K_X$ is nef. Numerical effectivity is weaker: an effective divisor in the numerical class need not supply a section of the actual line bundle. For pairs (X, Δ) , the corresponding divisor is $-(K_X + \Delta)$, the anti-log-canonical divisor. Lazić–Matsumura–Petrernell–Tsakanikas–Xie [22, Theorem A] established numerical nonvanishing for projective log canonical threefold pairs with nef anti-log-canonical divisor, assuming either that the variety is $\mathbb{Q}$ -factorial or that it has rational singularities. Müller [24, Corollary B] proved nonvanishing for projective klt threefold pairs with nef anti-log-canonical divisor. His higher-dimensional theorem assumes, in addition to nefness, that the anti-log-canonical divisor of the induced pair on the general fiber of the maximal rationally connected fibration is semiample [24, Theorem A].

Our principal conversion result uses smooth metric semipositivity and works in every dimension. Let X be smooth connected projective, let $L = -K_X$ carry a smooth semipositive Hermitian metric, and let P be any fixed pseudoeffective Cartier divisor. Then Theorem 5.1 gives the implication

$$H^0(X, m_j L - P) \neq 0 \text{ for unbounded positive } m_j \implies H^0(X, mL) \neq 0 \text{ for some } m > 0.$$

The fixed error P may itself have no effective representative. A single Iitaka base converts the effective divisors into a pseudoeffective rational divisor on that base. The controlled-boundary theorem makes this divisor rationally effective with correction on base primes whose pullbacks disappear over X . A rational function from the base then corrects the original rational section without introducing a pole on X .

Twisted differential forms supply the sequence through the determinant method of Lazić–Petrernell [23, Lemma 4.1], adapted in [22, Lemma 5.1]. Ou’s generic tangent-nefness theorem [30, Theorem 1.4], in the cotangent-subsheaf form [22, Theorem 4.1], makes the negative determinant

pseudoeffective. Our extraction argument starts with positive unbounded exponents and also includes numerically trivial L . Thus nonzero sections of $(\Omega_X^1)^{\otimes p} \otimes m_j L$ for one fixed $p \geq 0$ and unbounded positive m_j give an anticanonical section by Corollary 5.6. The same holds for exterior forms. This is an ordinary conversion; invariant sections require further control of characters.

1.3. Volume metrics and the distinct geometric refinements. Let X be smooth projective and let $L = -K_X$ carry a smooth semipositive metric. In a resolved rational fibration

$$X \xleftarrow{\pi} Z \xrightarrow{g} Y,$$

assume Z, Y are smooth projective, π is birational, and g is surjective. Write $K_{Z/X} = K_Z - \pi^* K_X$ for the relative canonical (Jacobian) divisor. Its canonical section belongs to $K_Z + \pi^* L$. When $\text{rank } g_* \mathcal{O}_Z(K_{Z/X}) = 1$, its squared fiberwise L^2 norm is a volume density ρ on the base. On the smooth base-change open where this section generates the adjoint line, $-\log \rho$ is a metric on $-K_Y$. Berndtsson's direct-image theory supplies its positivity [5, 6]. Extending it over the boundary with enough integrability is the geometric task preceding effectivity.

The main conversion proof uses a singular coefficient metric whose curvature dominates a positive base form and whose total-space multiplier ideal is trivial. Singular adjoint direct-image positivity of Păun–Takayama [31] then gives a metric with a global strict lower bound on $-K_Y + C$. Its integral boundary has exceptional pullback, and the distinguished section is locally square integrable.

Keeping the coefficient metric smooth gives a different output. The coarea formula and a primitive-representative curvature calculation produce an integral correction C with $\pi_* g^* C = 0$ and

$$|s_C|^2 e^{-\phi} \in L_{\text{loc}}^{1+\eta} \quad \text{for some } \eta > 0.$$

The metric is smooth and strictly positive on one ordinary open set. The higher integrability exponent permits a mixture with a globally strict metric while keeping s_C integrable. Thus the global bound needed for Theorem 1.1 follows from an explicit additional step.

The companion *Metric descent and rank-preserving contractions* supplies the toroidal-volume construction used here [29, Theorem 2.1]. For a normal projective model $p : W \rightarrow X$ and a surjective equidimensional toroidal map $f : W \rightarrow Y$, its theorem assumes adjoint rank one at degree zero on a smooth resolution and a patch, away from the Jacobian divisor, where the pulled-back curvature dominates a positive base form. It produces an effective rational correction A_0 with

$$p_* f^* A_0 = 0$$

and a semipositive metric on $-K_Y + A_0$ with unweighted locally integrable squared frame norms, smooth and strictly positive on a nonempty ordinary open set. That companion proves the toroidal Jacobian calculation, the all-valuation integrability and the boundary extension. We use its metric after checking the stated model, rank and curvature premises in each application. The rational correction is used directly in Corollary 5.4; no rounding or replacement of its unweighted metric by an integral-boundary hypothesis is needed.

For the invariant problem, Müller [24, Theorem C] obtains naturally invariant sections for projective sub-log-canonical pairs with semiample anti-log-canonical divisor and a commutative linear algebraic group action. Theorem 9.2 here assumes X smooth connected projective, $-K_X$ smoothly semipositive and $\chi(X, \mathcal{O}_X) \neq 0$. For any algebraic torus acting on X , it gives

$$H^0(X, -mK_X)^T \neq 0 \quad \text{for some } m > 0$$

with the natural tangent-determinant linearization. The companion invariant-index theorem, including its actual value at zero, supplies invariant cohomology [28, Theorem 1.1]; compact-averaged hard Lefschetz then gives forms with the coefficient shift $m + 1$. The local determinant and character arguments supply the sections needed here. The finite-cover structure and norm

used in the isometric application are those of the companion headline paper, [27, Theorem 2.1 and Lemma 2.2], which develops the geometric work of Demailly–Peternell–Schneider and Campana–Demailly–Peternell [16, 11]. Our algebraic boundary criterion, smooth coarea proof and correction ring are independent of that application.

Finally, a suitable auxiliary base has an effective big klt boundary Δ with $K_Y + \Delta \sim_{\mathbb{Q}} C$ for an effective rational C supported on divisors disappearing over X . When $\kappa(Y, C) = 0$, a log terminal model of the base is of Fano type: it admits an effective rational boundary Θ with klt singularities such that $-(K + \Theta)$ is ample. This is a statement about the auxiliary model. Its proof retains the precise contracted-prime support needed to return the resulting divisors to X .

The main route is the boundary and ring results followed by the single-Iitaka conversion in Section 5. Sections 6 and 7 retain the smooth coarea estimate and direct jet-separation proof of bigness; their applications reuse the common return of sections. Section 8 then develops the supported Fano-type model. The invariant base constructions in Sections 9 and 10 lead to compact-monodromy descent in Section 11. Finally, Section 12 constructs invariant tensors from extremal characters of the full equivariant Euler characteristic. That construction is independent of the weight-zero index input and descends tensors before applying ordinary conversion.

2. FROM AN INTEGRABLE SECTION TO A RATIONAL BOUNDARY

The analytic information in Theorem 1.1 concerns one section, rather than a nonvanishing local frame. The distinction already appears on a disk. For a weight $\phi = a \log |z|^2$ with $a \geq 0$ and a section $s_J = z^j$ with $j \in \mathbb{Z}_{\geq 0}$, the two integrability thresholds are

$$|s_J|^2 e^{-\phi} \in L_{\text{loc}}^1 \iff a < j + 1, \quad e^{-\phi} \in L_{\text{loc}}^1 \iff a < 1.$$

On a resolution, the Jacobian contributes an additional vanishing order. After dividing the fixed divisor by the multiple used for approximation, subtracting this Jacobian order and that of J will leave coefficients strictly below one. The resulting signed divisor has a positive part that can enter a klt boundary, while its negative part records the support on which a correction is permitted.

We use additive notation for rational line bundles. If a metric has local squared frame norm $e^{-\phi}$, its curvature is positive when ϕ is plurisubharmonic. A global strict lower bound means $\sqrt{-1}\partial\bar{\partial}\phi \geq c\omega$ for one Kähler form ω and one $c > 0$. Smooth changes of weight do not affect local integrability. The multiplier ideal $\mathcal{J}(\phi)$ consists of holomorphic germs u for which $|u|^2 e^{-\phi}$ is locally integrable. Section 4 uses the unweighted case $J = 0$; it first obtains a global strict bound while preserving integrability of squared frame norms.

Lemma 2.1 (Approximation retaining a distinguished section). *Let Y be smooth projective, let $J \geq 0$ be an integral divisor, and let B be a rational line bundle with a singular metric h having a global strict curvature lower bound. Suppose $\mathcal{J}(h) \supset \mathcal{O}_Y(-J)$. For every sufficiently large divisible integer r , the complete system $|rB|$ is nonempty. Let $\sigma : Y' \rightarrow Y$ be a log resolution of its base ideal, J , and the exceptional locus, and write*

$$\sigma^*|rB| = G + |\text{Mov}|,$$

where G is the fixed divisor and $|\text{Mov}|$ is base-point free. Then the simple-normal-crossing divisor

$$\Psi = G/r - \sigma^*J - K_{Y'/Y} \tag{2.1}$$

has every coefficient strictly less than one. Its negative part is supported on σ^*J and the exceptional primes.

Proof. Fix a very ample integral divisor H and put $d = \dim Y$. For r sufficiently large and divisible, the curvature of rh dominates the fixed smooth curvatures of $K_Y + iH$, simultaneously for $1 \leq i \leq d$. Nadel vanishing [25, 15] gives

$$H^i(Y, \mathcal{O}_Y(rB - iH) \otimes \mathcal{J}(r\phi)) = 0 \quad (1 \leq i \leq d).$$

Indeed, in adjoint form the coefficient line is $rB - K_Y - iH$, whose metric has a strict positive lower bound. Castelnuovo–Mumford regularity makes $\mathcal{O}_Y(rB) \otimes \mathcal{J}(r\phi)$ globally generated. This is a nonzero coherent sheaf of rank one. Consequently $|rB|$ is nonempty and its base ideal $\mathfrak{b}_r$ satisfies

$$\mathfrak{b}_r \supset \mathcal{J}(r\phi). \quad (2.2)$$

We next compare the fixed divisor with the singularity of h . Fix r and a prime S on Y' . At its general point choose a transverse coordinate t and write

$$a = \text{ord}_S G, \quad j = \text{ord}_S \sigma^* J, \quad k = \text{ord}_S K_{Y'/Y}.$$

On nested coordinate balls downstairs, the point-extension estimate of Ohsawa–Takegoshi, in its weighted Bergman formulation [26, 15], gives holomorphic functions u_z with

$$|u_z(z)|^2 = e^{r\phi(z)}, \quad \int |u_z|^2 e^{-r\phi} \leq C_r$$

for every center z in the smaller ball where $\phi(z) > -\infty$. The constant is independent of z ; it need not be independent of r . Since ϕ is bounded above on a slightly smaller ball, holomorphic mean-value estimates bound u_z uniformly there. Each germ of u_z belongs to $\mathcal{J}(r\phi)$, hence to $\mathfrak{b}_r$ by (2.2). Its pullback is therefore divisible by t^a near a general point of S . Cauchy estimates on nested polydisks bound $(u_z \circ \sigma)/t^a$ uniformly as well. Substituting $z = \sigma(w)$ proves

$$\phi \circ \sigma \leq \frac{a}{r} \log |t|^2 + O(1). \quad (2.3)$$

The assertion is automatic on the polar set of ϕ .

The density $|s_J|^2 e^{-\phi}$ is locally integrable downstairs. Change of variables contributes a factor of order $2k$ along S . By (2.3), its pullback is bounded below, up to a positive bounded factor, by $|t|^{2(j+k-a/r)}$ at a general point of S . Integrability in the transverse complex variable forces

$$\frac{a}{r} - j - k < 1.$$

These are exactly the coefficients in (2.1). Resolving the three supports makes that divisor simple normal crossings. Away from $\sigma^* J$ and the exceptional locus, its only possible contribution is $G/r \geq 0$, which proves the assertion about its negative part. $\square$

The strict inequalities are the essential output. A non-strict inequality would not produce a klt boundary. The local extension argument explains why integrability of the distinguished section is precisely the hypothesis that survives after subtracting $\sigma^* J$.

Theorem 2.2 (Weighted anticanonical approximation). *Let Y be smooth projective and let $J \geq 0$ be integral. Suppose $-K_Y + J$ has a singular metric with a global strict curvature lower bound and multiplier ideal containing $\mathcal{O}_Y(-J)$. Choose a sufficiently small ample rational divisor A . For a sufficiently large divisible integer r , resolve the complete system $|r(-K_Y + J - A)|$ and J by $\sigma : Y' \rightarrow Y$, and write*

$$\sigma^* |r(-K_Y + J - A)| = G + |\text{Mov}|.$$

Then $\Psi = G/r - \sigma^ J - K_{Y'/Y}$ has coefficients less than one, Ψ^+ is an effective klt boundary, and $D_0 = \Psi^-$ is effective and supported on $\sigma^* J$ and exceptional primes. Moreover*

$$D_0 \sim_{\mathbb{Q}} K_{Y'} + \Psi^+ + \text{Mov}/r + \sigma^* A. \quad (2.4)$$

The rational class $\text{Mov}/r + \sigma^ A$ is nef and big. If an algebraic torus acts on Y , J is invariant, and the indicated full systems are torus-stable, the resolution may be chosen equivariantly. Then G , $K_{Y'/Y}$, $\Psi^\pm$ and D_0 are invariant, and the free system $|\text{Mov}|$ is torus-stable. Identity (2.4) is an identity of ordinary rational line bundles; it does not assert that a chosen member of either the free system or the ample class is invariant.*

Proof. Subtracting a sufficiently small smooth ample metric preserves the strict lower bound and does not change the multiplier ideal. Apply Lemma 2.1 to $B = -K_Y + J - A$. Its identity $G + \text{Mov} \sim_{\mathbb{Q}} r\sigma^*B$ gives (2.4). The positive part of a simple normal crossing divisor with coefficients less than one is klt. The moving class is nef, and the pullback of the ample class A is nef and big, proving the last positivity assertion. In the equivariant setting the complete-system base ideal is invariant; equivariant resolution in characteristic zero, applied also to J , therefore supplies invariant fixed, relative canonical and boundary divisors. This does not require the original metric to be invariant. $\square$

The signed identity supplies more information than an effectivity conclusion. In particular, its negative part has an explicit support. It does not assert integrability of the norm of a nonvanishing frame; that stronger hypothesis enters Section 4.

3. EFFECTIVITY WITH A PRESCRIBED CORRECTION SUPPORT

The signed approximation separates two kinds of data: the negative part is confined to the permitted correction support, while the ample part can absorb a small multiple of any prescribed rational divisor. We use that ample part to insert the pseudoeffective divisor of Theorem 1.1. The resulting boundary is effective, big and klt, so big-boundary nonvanishing applies to its adjoint. We first record the passage from its real-linear conclusion to the rational linear equivalence required here.

Lemma 3.1 (Rational recovery). *Let D be a rational divisor on a normal projective variety. If D is real-linearly equivalent to an effective real divisor, then it is rationally linearly equivalent to an effective rational divisor.*

Proof. Write an effective real representative as $D + \sum_{j=1}^s a_j \text{div}(f_j)$, with $a_j \in \mathbb{R}$ and rational functions f_j . The supports of D and of these finitely many principal divisors give a finite system of rational linear inequalities asserting nonnegativity of every coefficient. The known real solution makes the resulting rational polyhedron nonempty. Such a polyhedron has a rational point: take a point in the relative interior of its smallest supporting face, solve that face's rational linear equations over $\mathbb{Q}$, and approximate within its rational affine space while preserving the remaining strict inequalities. Substituting that rational point gives the assertion. $\square$

Proof of Theorem 1.1. Fix a small ample rational divisor A whose smooth metric can be subtracted from the given metric on $-K_Y + C$ while retaining its global strict curvature lower bound. Apply Theorem 2.2 with $J = C$ and choose $r > 1$. On the resulting resolution $\sigma : Y' \rightarrow Y$, write $\Psi = \Psi^+ - \Psi^-$ and retain the free system $|\text{Mov}|$. The signed adjoint identity is

$$K_{Y'} + \Psi^+ + \text{Mov}/r + \sigma^*A \sim_{\mathbb{Q}} \Psi^-.$$

Choose a general $D_{\text{mov}} \in |\text{Mov}|$ meeting the resolved support transversely and without common components. All these data are fixed before choosing the pseudoeffective divisor M .

The ample cone is open, so $A + tM$ is ample for a sufficiently small positive rational t . Choose an effective rational representative $U_t \sim_{\mathbb{Q}} A + tM$ by dividing a general member of a sufficiently high very ample multiple. Bertini permits σ^*U_t and D_{mov} to meet the resolved support with simple normal crossings, with no common components; the new coefficients are less than one. Thus

$$\Delta_t = \Psi^+ + D_{\text{mov}}/r + \sigma^*U_t \tag{3.1}$$

is effective and klt. Its class is big because it contains the effective summand σ^*U_t , the pullback of an ample rational class. The adjoint identity becomes

$$K_{Y'} + \Delta_t \sim_{\mathbb{Q}} \Psi^- + t\sigma^*M. \tag{3.2}$$

Its right-hand side is pseudoeffective.

Big-boundary nonvanishing of Birkar–Cascini–Hacon–McKernan [8, Theorem D] therefore gives an effective real divisor real-linearly equivalent to $K_{Y'} + \Delta_t$. Lemma 3.1 makes this a rational linear equivalence with an effective rational divisor. Finally, $F = \sigma_* \Psi^-$ is effective and supported on C by Lemma 2.1. Push (3.2) down to Y and divide by t . The theorem follows with $C_0 = F/t$. $\square$

Proposition 3.2 (A fixed support correction before scaling). *Under the hypotheses of Theorem 1.1, there is an effective rational divisor F supported on C such that for every pseudoeffective rational divisor M , some positive rational t makes $F + tM$ rationally linearly equivalent to an effective rational divisor.*

Proof. In the preceding proof, the approximation and hence $F = \sigma_* \Psi^-$ were chosen independently of M . Only t and the general ample representative depended on M . $\square$

When $C = 0$, the correction vanishes. For nonzero C , it is the integrability of s_C that controls the support of Ψ^- ; no unweighted frame-integrability hypothesis has been imposed. The divisor to which nonvanishing is applied is the effective boundary (3.1).

3.1. Preserving integrability under a pseudoeffective perturbation. One can also insert M at the level of metrics, before approximation. This gives analytic information in addition to the effectivity proved above: a small pseudoeffective perturbation preserves the integrability of the distinguished section and the global strict curvature bound.

Lemma 3.3. *Under the hypotheses of Theorem 1.1, for every pseudoeffective rational divisor M there is a positive rational ϵ such that $-K_Y + C + \epsilon M$ has a metric with a global strict curvature lower bound and multiplier ideal containing $\mathcal{O}_Y(-C)$.*

Proof. Write ϕ for the given metric and choose a semipositive singular metric with weights θ on M . Strong openness [19] gives a common $\tau > 0$ on a finite relatively compact coordinate covering such that $|s_C|^2 e^{-(1+\tau)\phi}$ is locally integrable. Put $q = (1 + \tau)/\tau$. Hölder’s inequality with respect to the measure $|s_C|^2 d\lambda$ gives

$$\int |s_C|^2 e^{-\phi - \epsilon\theta} \leq \left(\int |s_C|^2 e^{-(1+\tau)\phi} \right)^{1/(1+\tau)} \left(\int |s_C|^2 e^{-q\epsilon\theta} \right)^{1/q}.$$

Small-exponent integrability of plurisubharmonic functions [33] makes the second factor finite for sufficiently small positive rational ϵ . The finite covering permits one choice of ϵ . The weight $\phi + \epsilon\theta$ retains the global strict curvature lower bound because the metric on M is semipositive. $\square$

Subtract a sufficiently small ample rational A from the perturbed metric, retaining its strict lower bound. Applying Lemma 2.1 to $-K_Y + C + \epsilon M - A$ with $r > 1$ then gives a second boundary construction. For its signed divisor Ψ and a general free member D_{mov} , add a general small-coefficient representative $U \sim_{\mathbb{Q}} A$. Then $\Delta = \Psi^+ + D_{\text{mov}}/r + \sigma^* U$ is effective, big and klt, and $K_{Y'} + \Delta \sim_{\mathbb{Q}} \epsilon \sigma^* M + \Psi^-$. Nonvanishing, rational recovery and pushforward apply as before. Here the approximation itself depends on the metric on M ; the support conclusion is unchanged.

4. UNWEIGHTED INTEGRABILITY AND NEARBY ADJOINT RINGS

The criterion just proved accepts a weighted integral and a global strict curvature bound. For finite generation we instead begin with integrable squared *frame* norms and strict positivity on one ordinary open set. This permits perturbation in arbitrary finitely many rational directions. We first make the passage from local strictness to a global lower bound while preserving integrability.

Lemma 4.1 (An integrable neighborhood). *Let Y be smooth projective of positive dimension and let B be a rational line bundle. Suppose B has a semipositive singular metric with locally integrable squared frame norms, smooth and strictly positive on a nonempty ordinary open set. For any finite list of rational line bundles $P_1, \dots, P_s$, there are $\epsilon \in \mathbb{Q}_{>0}$ and a common ample rational*

line bundle H' such that each $B + \epsilon P_i - H'$ is rationally linearly equivalent to an effective divisor Δ_i with (Y, Δ_i) klt.

Proof. The absolutely continuous part of the curvature current has positive top-degree mass on the given open set. Boucksom's volume criterion [9, Theorem 1.2(i)] implies that B is big. A rational big decomposition therefore gives a metric with algebraic singularities and weights ψ whose curvature dominates $c\omega$ for some $c > 0$. Write ϕ for the original weights. Strong openness and compactness give $\eta > 0$ with $e^{-(1+\eta)\phi}$ locally integrable on one finite covering. Small-exponent integrability supplies $a > 0$ with $e^{-a\psi}$ locally integrable on that covering as well.

Choose $p > 1$ with $p < 1 + \eta$ and let $q = p/(p - 1)$. For small positive rational u we have $p(1 - u) < 1 + \eta$ and $qu < a$. Hölder's inequality then makes

$$e^{-((1-u)\phi + u\psi)}$$

locally integrable. The mixed metric has curvature at least $u\omega$. This is the required global strict lower bound; positivity on the original open set was used only to establish bigness.

Choose smooth metrics on all P_i and on one ample integral divisor H . Their curvatures have uniformly bounded norm relative to ω . Taking rational $\epsilon > 0$ and $h > 0$ sufficiently small, subtracting $H' = hH$ and adding ϵP_i leaves every mixed metric with a common strict lower bound. Smooth changes of weights preserve integrability.

It remains to turn a rational line bundle D with such a strict integrable metric into an effective klt divisor. Apply Lemma 2.1 with $J = 0$ and $r > 1$ to D . Take a general member of $|rD|$ and denote its divisor divided by r by Δ . On the resolution its pullback is $G/r + D_{\text{mov}}/r$. The lemma says that $G/r - K_{Y'/Y}$ has coefficients less than one. A general free member meets the fixed support transversely and has coefficient $1/r < 1$. Thus all coefficients of $\sigma^*\Delta - K_{Y'/Y}$ are less than one with simple normal crossing support. This is exactly the klt condition for the effective pair (Y, Δ) . Since $\Delta \sim_{\mathbb{Q}} D$, apply this argument to $D = B + \epsilon P_i - H'$ for each i . $\square$

Corollary 4.2. *Under the hypotheses of Lemma 4.1, B has an effective big klt rational representative. The same holds for $B + \epsilon P_i$ for the sufficiently small simultaneous perturbations in that lemma.*

Proof. Add to Δ_i a general effective rational representative of H' with sufficiently small coefficients. On a common log resolution, Bertini preserves the klt inequalities. The sum is big because it contains an effective representative of the ample class H' . For B itself, take the sole direction $P_1 = 0$. $\square$

Theorem 4.3 (Finite generation near a correction divisor). *Let Y be smooth projective of positive dimension, and let A_0 be a rational divisor. Suppose $-K_Y + A_0$ has a semipositive singular metric with locally integrable squared frame norms, smooth and strictly positive on a nonempty ordinary open set. Given rational divisors $P_1, \dots, P_s$, there are $\epsilon \in \mathbb{Q}_{>0}$ and an integer $k_0 > 0$ such that the actual divisors $B_i = k_0(A_0 + \epsilon P_i)$ are Cartier and*

$$R(Y; B_1, \dots, B_s) = \bigoplus_{(a_1, \dots, a_s) \in \mathbb{N}^s} H^0\left(Y, \mathcal{O}_Y\left(\sum_i a_i B_i\right)\right)$$

is a finitely generated $\mathbb{C}$ -algebra.

Proof. Lemma 4.1 gives effective klt rational boundaries Δ_i and one ample rational H' such that

$$\Delta_i \sim_{\mathbb{Q}} -K_Y + A_0 + \epsilon P_i - H'.$$

The multigraded adjoint finite-generation theorem of Corti–Lazić [14, Theorem 3.2(1)], based on Cascini–Lazić [13], applies to the divisors $K_Y + \Delta_i + H'$. Clear all denominators and all rational linear equivalences with a common integer k_0 . Multiplication by the corresponding rational

functions, compatibly in each multidegree, identifies a multigraded Veronese of that adjoint ring with the displayed ring. Finite-index multigraded Veronese subrings are finitely generated [14, Lemma 3.1]. The construction uses rational linear equivalence and actual Cartier divisors, not only their numerical classes. $\square$

4.1. Direct selection of an integrable divisor. There is a second way to perform the last step of Lemma 4.1. It chooses a section by averaging negative powers of its norm, instead of using a general member on a log resolution. This gives a direct analytic description of the effective klt representative.

Let D be a rational line bundle with a globally strictly positive metric of weights ϕ_D , and suppose $e^{-\phi_D}$ is locally integrable. For a large divisible integer $m > 1$, take an orthonormal basis $s_1, \dots, s_N$ of the global sections of mD that are square integrable with weight $m\phi_D$ and a fixed smooth volume form. The global Bergman estimate is

$$\frac{1}{m} \log \sum_{j=1}^N |s_j|^2 \geq \phi_D - C_m \quad (4.1)$$

on each relatively compact coordinate patch. We recall the construction because this version uses global sections of mD without a fixed auxiliary twist.

Fix a point y where $\phi_D(y)$ is finite. Point extension on a coordinate ball gives a local holomorphic section taking squared value $e^{m\phi_D(y)}$ at y , with uniformly bounded weighted norm. Multiply it by a cutoff equal to one on a smaller ball and solve the resulting $\bar{\partial}$ equation. In top holomorphic degree the coefficient line is $mD - K_Y$. Add a cutoff logarithmic pole $d \log |z - y|^2$ to the estimating weight, where $d = \dim Y$. For sufficiently large m , the strict curvature of mD absorbs both the fixed canonical curvature and the negative curvature of the cutoff pole, uniformly for y in a smaller ball. The correction is holomorphic near y ; its integrability against $|z - y|^{-2d}$ forces its value there to vanish. The corrected global section retains the specified value and has uniformly bounded $m\phi_D$ -weighted norm.

For the singular-weight solution, first choose rational trivializing sections of an integral multiple of D and of K_Y . Enlarge the support of their zeros and poles by an effective divisor so that the resulting divisor is ample. Its complement is affine and trivializes the required integral line bundles. Use Stein exhaustions of that complement, approximate the psh weights on relatively compact exhaustion sets and pass to weak limits. In bidegree $(d, 1)$ the L^2 estimate uses contraction with the inverse positive line-curvature matrix; its bound is independent of an auxiliary complete Kähler metric chosen to dominate a fixed reference metric. The estimates therefore persist through the exhaustion. Since the original psh weights are locally bounded above, weighted square integrability implies ordinary square integrability near the removed divisor. The resulting holomorphic section extends across it. The local weighted estimate is Demailly's Bergman approximation [15, Section 1, Theorem 1.2], with the point-extension input of Ohsawa–Takegoshi [26]. The strict global curvature is what permits the untwisted global construction just given. Expanding the constructed section in the orthonormal basis proves (4.1); the estimate is automatic on the polar set of the weight.

It follows that $(\sum_j |s_j|^2)^{-1/m}$ is locally integrable. Average over the unit sphere in $\mathbb{C}^N$. Unitary invariance gives

$$\int_{\|a\|=1} \left| \sum_{j=1}^N a_j s_j(y) \right|^{-2/m} da = c_{m,N} \left(\sum_{j=1}^N |s_j(y)|^2 \right)^{-1/m}.$$

For $N \geq 2$, the possible singularity is an inverse power of one complex coordinate and $c_{m,N} < \infty$ because $m > 1$. For $N = 1$ the identity is immediate. Fubini on a finite covering gives a nonzero section s of mD with $|s|^{-2/m}$ locally integrable. On a log resolution, change of variables identifies

this condition with all discrepancy coefficients of $\text{div}(s)/m$ being less than one. Thus $\text{div}(s)/m$ is the required effective klt rational representative of D .

5. REMOVING A FIXED PSEUDOEFFECTIVE ERROR

The controlled-boundary theorem becomes useful for anticanonical nonvanishing when the boundary on an auxiliary base disappears after pullback and birational pushforward. We now construct precisely such a base. The error in the following theorem is an arbitrary fixed pseudoeffective divisor; it need not be effective.

Theorem 5.1. *Let X be a smooth connected projective complex variety and let $L = -K_X$ have a smooth semipositive Hermitian metric. Let P be a pseudoeffective Cartier divisor. If there are effective integral $N_j \sim m_j L - P$ for unbounded positive integers m_j , then $H^0(X, mL) \neq 0$ for some positive integer m .*

We use the Iitaka fibration of just one N_j . Ratios of its sections force all subsequent divisors to vary affinely on the generic fiber. The vertical slopes define a pseudoeffective divisor on the base. A rank-one adjoint direct image supplies the metric required by Theorem 1.1, and its boundary is exceptional over X . Thus the correction allowed by that theorem causes no poles in the section returned to X .

5.1. Vertical minima and return of sections. We record the return step once, including the exceptional divisors that can appear when a smooth model ceases to be equidimensional.

Lemma 5.2. *Let $\pi : Z \rightarrow X$ be a projective birational morphism of smooth projective varieties and let $f : Z \rightarrow Y$ be a surjective morphism to a smooth projective variety with connected general fibers, and let L be a rational line bundle on X . Assume every prime of Z mapping into codimension at least two in Y is π -exceptional. Let $R \sim_{\mathbb{Q}} a\pi^*L$, $a > 0$, be a rational divisor with nonnegative horizontal part. For each base prime Q put*

$$\mu_Q(R) = \min_{D \rightarrow Q} \frac{\text{ord}_D R}{\text{ord}_D f^*Q}, \quad M = \sum_Q \mu_Q(R)Q.$$

Suppose M is pseudoeffective. If there is an effective rational C_0 , all components of whose pullback are π -exceptional, such that $M + C_0 \sim_{\mathbb{Q}} G \geq 0$, then L is rationally linearly equivalent to an effective rational divisor. The same conclusion holds if the premise is $\epsilon M + C_0 \sim_{\mathbb{Q}} G \geq 0$ for some rational $\epsilon > 0$.

For the invariant conclusion, give L a specified linearization and assume

$$R = c \text{div}(\pi^*t), \quad cr = a,$$

where $c > 0$ is rational and t is a nonzero invariant rational section of the specified linearized L^r , with $r > 0$ an integer for which L^r is a line bundle. Assume also that every function pulled back from Y is invariant in $\mathbb{C}(X)$. Then the resulting section of a positive multiple of L is invariant for that linearization; no regular group action on Z is required.

Proof. Only finitely many coefficients of M are nonzero. Replace C_0, G by $C_0/\epsilon, G/\epsilon$ in the last variant. The divisor

$$R + f^*(G - M - C_0) \sim_{\mathbb{Q}} a\pi^*L$$

has nonnegative horizontal coefficients. At a prime dominating Q outside the support of C_0 , this follows from the minimum defining M and from $G \geq 0$. All other possibly negative primes are π -exceptional, either by the support premise on C_0 or by the assumption on primes mapping into codimension two. Its pushforward is therefore effective and rationally linearly equivalent to aL . Clearing denominators proves the assertion. In the invariant case choose the clearing integer n also so that nc is integral. The operation multiplies t^{nc} , a section of $L^{ncr} = L^{na}$, by a rational

function pulled from Y . Both factors are invariant, and extension across codimension two on the smooth variety X preserves invariance. $\square$

The required model condition follows from flattening, rather than from an assertion that a final smooth resolution is flat. Starting with a resolved rational map and its Stein factorization, flatten over a modification of the base [32, Theorem 5.2.2]. Normalize the main component, resolve the base, flatten again if necessary, and then resolve the source. On the intervening flat model a prime cannot map into base codimension two: its inverse image has dimension at most $\dim X - 2$. Any prime on the final resolution with that property is consequently exceptional over the intervening model, and hence over X .

5.2. The Iitaka base and its strict metric.

Proof of Theorem 5.1. Pass to a subsequence with $m_0 < m_1 < m_2 < \dots$, and take the Iitaka fibration of N_1 . Resolve a system of maximal image dimension, take its Stein factorization, and prepare the model by flattening as above:

$$X \xleftarrow{\pi} Z \xrightarrow{f} Y.$$

The base field $\mathbb{C}(Y)$ is relatively algebraically closed in $\mathbb{C}(X)$ and contains every ratio of sections of every positive multiple of N_1 . Indeed, a ratio transcendental over the original image field would increase the image dimension after multiplication of sections into a common system. All other ratios lie in its relative algebraic closure, which is the Stein field. If $\dim Y > 0$, we may choose an ample H on Y and an integer $b > 0$ with

$$b\pi^*N_1 \sim f^*H + D_1, \quad D_1 \geq 0. \quad (5.1)$$

The polarization of the original image pulls back to a big divisor on Y ; a sufficiently large multiple dominates H , which proves this assertion on the modified base.

Put $R = (N_1 - N_0)/(m_1 - m_0) \sim_{\mathbb{Q}} L$. For $j > 1$ the two effective divisors

$$(m_j - m_1)N_0 + (m_1 - m_0)N_j, \quad (m_j - m_0)N_1$$

are linearly equivalent. Their ratio is a base function. Consequently there are rational principal divisors $Q_j \sim_{\mathbb{Q}} 0$ on Y with the exact identities

$$\pi^*N_j = \pi^*N_0 + (m_j - m_0)(\pi^*R + f^*Q_j). \quad (5.2)$$

Effectivity as $m_j \rightarrow \infty$ makes the horizontal part of π^*R nonnegative. If Y is a point, this already proves the theorem. If $\dim Y = \dim X$, then N_1 is big, so $m_1L \sim N_1 + P$ is big and again has a section in a positive multiple. We henceforth assume $0 < \dim Y < \dim X$.

Define $M = \sum_Q \mu_Q(\pi^*R)Q$ as in Lemma 5.2. There is a fixed effective rational J on Y such that

$$M + Q_j + \frac{J}{m_j - m_0} \geq 0.$$

For example its coefficient at Q can be the maximum of $\text{ord}_D \pi^*N_0 / \text{ord}_D f^*Q$ over $D \mapsto Q$. These maxima have finite support. The inequality follows by taking each coefficient in (5.2). Since Q_j is rational principal, the numerical classes of these effective divisors converge to $[M]$. Thus M is pseudoeffective.

The remaining problem is to make M effective with a correction exceptional over X . Write $E = K_{Z/X} \geq 0$. We first show

$$\text{rank } f_*\mathcal{O}_Z(E) = 1. \quad (5.3)$$

The Jacobian section supplies rank at least one. If the rank were larger, twisting by kH for large k would supply two sections of $E + kf^*H$ independent on the generic fiber. Multiply by the section of $bk\pi^*N_1 - kf^*H$ furnished by (5.1). The resulting ratio belongs to $|E + bk\pi^*N_1|$. Since $\pi_*\mathcal{O}_Z(E) = \mathcal{O}_X$, it is a ratio from a multiple of N_1 , and hence a base function, a contradiction.

The equality of pushforwards used here follows from normality of X : a rational function with possible poles only on exceptional divisors has no pole at any prime of X and extends across codimension two.

Write

$$(f_*\mathcal{O}_Z(E))^{**} = \mathcal{O}_Y(C), \quad C \geq 0,$$

where C is the divisor of the canonical section. The divisor C is integral. The torsion-free rank-one direct image agrees with its hull at every codimension-one point. If $Q \subset \text{Supp } C$, dividing the canonical section by a local equation of Q is therefore allowed at its generic point. Hence every prime of Z dominating Q occurs in E . The model condition handles components over higher codimension, so every component of f^*C is exceptional over X .

We now verify all analytic premises of Theorem 1.1. Combining (5.1) with $N_1 \sim m_1 L - P$ gives

$$bm_1\pi^*L \sim b\pi^*P + f^*H + D_1.$$

A positive-current metric for P , an ample metric for H , and the divisor metric of D_1 therefore give a singular metric on π^*L whose curvature dominates a positive multiple of $f^*\omega_Y$. Mix a sufficiently small positive fraction of its weight with the smooth semipositive pullback metric. Skoda's small-exponent integrability theorem, on a finite cover of the compact Z , gives a metric h satisfying

$$\Theta_h \geq \delta f^*\omega_Y, \quad \mathcal{J}(h) = \mathcal{O}_Z, \quad \delta > 0. \quad (5.4)$$

The metric h need not be smooth. The triviality of its multiplier ideal is a statement on the whole total space; a uniform claim on all fibers is neither needed nor asserted.

Apply singular adjoint direct-image positivity [31, Theorem 3.3.5]; see also [20, Theorem 21.1 and Corollary 21.2]. The natural inclusion of the multiplier-ideal direct image into the ordinary adjoint direct image is an equality here, because $\mathcal{J}(h) = \mathcal{O}_Z$ in (5.4). In particular the generic-isomorphism hypothesis in the cited theorem holds, and it gives a metric on the ordinary sheaf. Projection formula and (5.3) identify its rank-one hull as

$$(f_*\mathcal{O}_Z(K_{Z/Y} + \pi^*L))^{**} = \mathcal{O}_Y(-K_Y + C).$$

It acquires a metric g with $\Theta_g \geq \delta\omega_Y$. For the lower bound, subtract a local potential of $\delta f^*\omega_Y$ upstairs; the fiber integral subtracts the corresponding base potential downstairs. The direct-image theorem extends the metric across divisors; plurisubharmonic extension handles the remaining codimension-two set. Thus this is a global strict curvature bound on the actual line bundle, including its divisorial correction.

The distinguished section has finite norm. On the smooth base-change locus its norm times coordinate base volume is the pushforward of the squared norm of the Jacobian section of $K_Z + \pi^*L$. The latter is locally integrable by $\mathcal{J}(h) = \mathcal{O}_Z$. Properness and Fubini give finite local mass on the base; the omitted analytic subsets have measure zero. In the line $-K_Y + C$ the distinguished local section is s_C . Consequently

$$\mathcal{J}(g) \supset \mathcal{O}_Y(-C).$$

We have verified global strictness, the precise multiplier-ideal inclusion, and exceptional pullback of the integral boundary. Theorem 1.1 gives $M + C_0 \sim_{\mathbb{Q}} G \geq 0$, with $C_0 \geq 0$ supported on C . Lemma 5.2 now gives the required section on X . $\square$

5.3. The smooth toroidal metric for the same base. A toroidal embedding is locally modeled on a toric variety with its dense torus; its boundary corresponds to the toric boundary. We use strict toroidal embeddings, whose boundary components are normal. A toroidal morphism is locally a toric morphism, described on the dense tori by monomials. This local description keeps track of the exact ramification exponents in the metric correction.

There is another way to supply the metric on the Iitaka base. It keeps the original smooth anticanonical metric throughout fiber integration and gives unweighted integrability. We give the geometric verification because the two metric constructions have different conclusions.

Lemma 5.3. *Let Z be smooth projective of dimension n , and let θ, η be smooth semipositive closed $(1, 1)$ -forms. Suppose $a[\theta] - [\eta]$ is pseudoeffective for some $a > 0$. If ν is the maximum rank of θ , then $\ker \theta \subset \ker \eta$ on its rank- ν locus. If $\eta = f^*\omega_Y$ on an ordinary open meeting the rank- ν locus, with f submersive and $\omega_Y > 0$, there is a smaller nonempty patch on which $\theta \geq cf^*\omega_Y$ for some $c > 0$.*

Proof. There is nothing to prove when $\nu = n$. Otherwise choose a Kähler form ω . Every mixed intersection of θ and η of total degree $\nu + 1$, completed by $\omega^{n-\nu-1}$, is nonnegative. Replacing one η at a time by $a\theta$ can only increase the intersection, because a pseudoeffective class intersects nonnegatively with $n - 1$ nef classes. The resulting upper bound is a multiple of $\int_Z \theta^{\nu+1} \omega^{n-\nu-1} = 0$. Thus $(\theta + \eta)^{\nu+1} = 0$ pointwise, by semipositivity. On the rank- ν locus $\theta + \eta$ has the same kernel as θ ; this is the asserted inclusion. On a relatively compact constant-rank patch, the nonzero eigenvalues of θ have a positive lower bound and $f^*\omega_Y$ is bounded, giving the final inequality. $\square$

We also isolate the algebraic consequence of unweighted integrability.

Corollary 5.4. *Let Y be smooth projective of positive dimension and $A_0 \geq 0$ rational. Suppose $B = -K_Y + A_0$ has a semipositive metric with locally integrable squared frame norms, smooth and strictly positive on a nonempty ordinary open. For every pseudoeffective rational M there is a rational $\epsilon > 0$ such that $A_0 + \epsilon M$ is rationally linearly equivalent to an effective rational divisor.*

Proof. Apply Lemma 4.1 in the direction M . It gives an ample rational H' , a sufficiently small rational $\epsilon > 0$, and an effective klt boundary $\Delta_0 \sim_{\mathbb{Q}} B + \epsilon M - H'$. Represent H' by a general effective rational divisor with small coefficients transverse to a log resolution of Δ_0 . Its addition gives an effective big klt boundary $\Delta \sim_{\mathbb{Q}} B + \epsilon M$. The log canonical divisor $K_Y + \Delta \sim_{\mathbb{Q}} A_0 + \epsilon M$ is pseudoeffective. Big-boundary nonvanishing [8, Theorem D] applies. Lemma 3.1 gives rational linear effectivity. $\square$

Prepare the rational Iitaka map birationally as an equidimensional strict toroidal morphism $f : W \rightarrow Y$, with W normal, Y smooth, and $p : W \rightarrow X$ birational. Such a preparation follows from weak toroidalization [1, Theorem 1.1] and equidimensional toroidal subdivision [2, Proposition 4.4]. Mark the nonisomorphism locus in the source boundary before these operations. No finite base alteration or reduced-fiber assertion is used here. Take a smooth resolution $Z \rightarrow W$, with maps $\pi : Z \rightarrow X$ and $g : Z \rightarrow Y$. The resolution itself need not be equidimensional. The generic rank-one argument (5.3) is unchanged. Likewise the horizontal interpolation and the vertical minima may be computed on W , where every vertical prime dominates a base prime.

Equation (5.1), pulled to Z , says that a multiple of $[\Theta_{\pi^*L}]$ dominates the pullback of a positive base class pseudoeffectively. Lemma 5.3 gives a horizontal strictness patch above the smooth base-change locus, away from the Jacobian divisor. The Jacobian section is nonzero there. The model, rank-one, and strict-patch hypotheses have now been verified for the original smooth coefficient metric. These are the hypotheses of the toroidal-volume theorem in [29, Theorem 2.1]. That theorem supplies a rational $A_0 \geq 0$ with $p_*f^*A_0 = 0$ and the unweighted integrable metric of Corollary 5.4. Its coefficients are

$$\text{ord}_Q A_0 = \max \left\{ 0, \min_{D \rightarrow Q} \left(\frac{1 + \text{ord}_D K_{W/X}}{\text{ord}_D f^*Q} - 1 \right) \right\}.$$

The all-valuation integrability and extension assertions here are part of that theorem, rather than consequences of generic rank one alone. Corollary 5.4 and Lemma 5.2, on a common resolution, complete the same conversion. The output here is unweighted integrability for a rational correction:

a nonvanishing frame has finite norm. The smooth coarea construction in Section 6 instead gives an integral correction and an $L^{1+\eta}$ bound for the distinguished section. Lemma 6.5 converts that output to the same supported effectivity premise in Lemma 5.2.

5.4. The field of all pseudoeffectively dominated systems. A second base construction uses every effective system dominated by a multiple of L . It is useful when no distinguished divisor should be chosen in advance. We retain the different maximality argument and the explicit integral correction it produces.

Call a subsystem of an integral divisor D admissible if it has a nonzero section and $bL - D$ is pseudoeffective for some integer $b > 0$. Choose finitely many admissible systems whose joint image has maximum dimension d . Every further admissible ratio is algebraic over this image field. Take its relative algebraic closure and a smooth projective connected-fiber model $\pi : Z \rightarrow X$, $f : Z \rightarrow Y$, prepared by flattening as before. For every ample H on Y there is a $b_0 > 0$ with

$$b_0 \pi^* L - f^* H \text{ pseudoeffective.} \quad (5.5)$$

Indeed the sum of the chosen divisors dominates the pullback of the product polarization; its restriction to the joint image is big. The sum itself is pseudoeffectively dominated by a multiple of L , and a large multiple of the product polarization dominates H . If $d = \dim X$, this makes L big.

Take sections u_j defining $N_j \sim m_j L - P$, put $r = m_2 - m_1$, and let $t = \pi^*(u_2/u_1)$, a rational section of $r\pi^* L$. For $j > 2$ the expressions $u_j u_1^{m_j - m_2}$ and $u_2^{m_j - m_1}$ belong to the same admissible system, namely a multiple of $m_2 L - P$. Writing $D_t = \text{div}(t)$ gives

$$r\pi^* N_j = r\pi^* N_1 + (m_j - m_1) D_t + f^* \text{div}(g_j), \quad g_j \in \mathbb{C}(Y)^*.$$

Thus D_t has nonnegative horizontal part, and its divisor of vertical minima M is pseudoeffective by the same fixed-error inequality used in (5.2). The case $d = 0$ is already settled.

For $d > 0$, the full adjoint sheaf $f_* \mathcal{O}_Z(E)$, $E = K_{Z/X}$, has rank one. Otherwise two generic-fiber independent sections can be made global in $E + kf^* H$. After pushforward their effective divisors give a subsystem of $k\pi_* f^* H$ on X whose ratio is not in $\mathbb{C}(Y)$. By (5.5) this system is admissible, contradicting maximality. Mixing a positive-current metric furnished by (5.5) with the smooth pullback metric again gives a globally defined coefficient h with $\Theta_h \geq \delta f^* \omega_Y$ and $\mathcal{J}(h) = \mathcal{O}_Z$. The singular direct-image theorem therefore applies to the full actual sheaf as in the first proof. Let w be the pushed-forward Jacobian volume density on its smooth base-change locus. Then $-\log w$ has curvature at least $\delta \omega_Y$ there, and w has finite local mass.

Here the boundary can be written directly. Set

$$c_Q = \min_{D \mapsto Q} \left\lfloor \frac{\text{ord}_D E}{\text{ord}_D f^* Q} \right\rfloor, \quad C = \sum_Q c_Q Q.$$

It is effective and integral. If a prime over Q is nonexceptional then its coefficient in E is zero, so $c_Q = 0$. Hence $f^* C$ is exceptional, including the primes over codimension two by the model condition. Near a general point of a prime D attaining the minimum, write $f = (z_1^e, z_2, \dots, z_d)$ and $a = \text{ord}_D E$. A plurisubharmonic coefficient weight is bounded above on a relatively compact chart. Its metric norm is therefore bounded below by a positive constant times a smooth norm. Integration of this chart gives

$$w(y) \geq c |y_1|^{2(a+1-e)/e}.$$

If $a = qe + s$, $0 \leq s < e$, then $(a + 1 - e)/e = q + (s + 1 - e)/e \leq q = \lfloor a/e \rfloor$. It follows that $\psi = -\log w + \log |s_C|^2$ is locally bounded above at general points of every missing divisor. Extend plurisubharmonically across these divisors and then across codimension two, subtracting local potentials of $\delta \omega_Y$ to preserve the lower bound. This gives a metric on $-K_Y + C$ with global strict curvature and

$$|s_C|^2 e^{-\psi} = w \in L_{\text{loc}}^1.$$

Theorem 1.1 applies to M . Lemma 5.2, now with the rational section t , completes the alternative construction. Thus its distinct ingredients are the field of all admissible systems, full rank one from maximality, and the explicit floor correction; the algebraic return is unchanged.

5.5. From twisted tensors to a fixed error. The determinant method of Lazić–Paternell [23, Lemma 4.1], and its formulation in [22, Lemma 5.1], provides the link to twisted differentials. Those cited extraction statements include a numerical-nontriviality hypothesis. The following direct construction also covers a numerically trivial L , because the starting exponents here are positive and unbounded.

Lemma 5.5. *Let X be smooth connected projective, let $L = -K_X$ be nef, and fix $p > 0$. If $H^0(X, (\Omega_X^1)^{\otimes p} \otimes mL) \neq 0$ for unbounded positive integers m , there are a pseudoeffective Cartier divisor P and effective integral $N_j \sim m_j L - P$ for unbounded positive m_j . If the starting sections are invariant for a torus and the induced linearizations, the determinant sections defining N_j are invariant.*

Proof. Each section defines a map $\mathcal{O}_X(-mL) \rightarrow (\Omega_X^1)^{\otimes p}$. Saturate their generic span, obtaining a coherent subsheaf $\mathcal{S}$. Finitely many maps already span at the generic point, and saturation ensures every original map factors through $\mathcal{S}$. Let $a = \text{rank } \mathcal{S}$. The reflexive determinant $A = (\wedge^a \mathcal{S})^{**}$ is a line bundle on smooth X . The cotangent-subsheaf consequence of generic tangent nefness [22, Theorem 4.1] gives $-A$ pseudoeffective. More explicitly, saturate A in the corresponding exterior power of the tensor bundle and embed that exterior power in a cotangent tensor power. The negative of the saturated line is pseudoeffective; passing back to $-A$ adds an effective divisor. Put $P = -A$.

Choose a generic basis from finitely many starting sections. Every nonzero section of arbitrarily large twist can replace one vector of that basis while keeping a basis. On an unbounded subsequence the same slot works. Wedge it with the remaining fixed vectors. This gives nonzero global sections of $A \otimes m_j L$, where m_j is the varying exponent plus a fixed sum of exponents. The sums remain positive and unbounded. The wedges extend globally by the determinant map and reflexive extension across codimension two. If the maps are equivariant, their saturation and determinant carry the induced action, and the wedge of invariant sections is invariant. $\square$

Corollary 5.6. *Under the metric hypotheses of Theorem 5.1, if $H^0(X, (\Omega_X^1)^{\otimes p} \otimes mL) \neq 0$ for a fixed $p \geq 0$ and unbounded positive m , then a positive multiple of L has a section. The same holds with Ω_X^p in place of the tensor power.*

Proof. For $p = 0$ a section is already given. Otherwise apply Lemma 5.5 and Theorem 5.1. Exterior forms embed in tensor powers by alternation in characteristic zero. This conversion is an ordinary statement; invariant determinant sections alone do not turn its conclusion into an invariant section. $\square$

5.6. A movable-slope proof of the determinant input. For completeness we give a second proof of the pseudoeffectivity used in Lemma 5.5, under its smooth-metric application hypotheses. This proof explains the ramification correction in the relative-volume argument. It uses movable slope theory [18], duality between pseudoeffective divisors and movable curves [10], and the positive-minimal-slope algebraicity theorem of Campana–Păun [12, Theorem 1.1].

For a torsion-free sheaf $\mathcal{E}$ and movable curve class α , its slope is $\mu_\alpha(\mathcal{E}) = c_1(\mathcal{E}) \cdot \alpha / \text{rank } \mathcal{E}$. The symbols $\mu_{\max, \alpha}$ and $\mu_{\min, \alpha}$ denote the greatest and least slopes in its Harder–Narasimhan filtration, whose successive quotients are semistable with decreasing slopes.

Suppose a saturated subsheaf of a cotangent tensor power on smooth projective X had determinant of positive degree against a movable curve class α . Tensor-product slope inequalities imply

$\mu_{\max, \alpha}(\Omega_X^1) > 0$, equivalently $\mu_{\min, \alpha}(T_X) < 0$. Since $c_1(T_X) \cdot \alpha \geq 0$, the positive-slope part $\mathcal{F}$ of the Harder–Narasimhan filtration of T_X satisfies

$$\mu_{\min, \alpha}(\mathcal{F}) > 0, \quad \mu_{\max, \alpha}(T_X/\mathcal{F}) \leq 0, \quad c_1(T_X/\mathcal{F}) \cdot \alpha < 0.$$

The bracket obstruction from $\wedge^2 \mathcal{F}$ to $T_X/\mathcal{F}$ vanishes by the slope inequality. Thus $\mathcal{F}$ is a foliation, and the cited algebraicity theorem realizes it as the relative tangent foliation of a rational map with connected general fibers. Prepare an equidimensional model of that map, then take a smooth resolution with maps $\mu : W \rightarrow X$ and $f : W \rightarrow Y$. Put $L = -K_X$ and retain its smooth semipositive metric.

The relative adjoint bundle $K_{W/Y} + \mu^*L$ has a nonzero section on a general fiber: there it is represented by the effective Jacobian $K_{W/X}$. On the smooth bundle and base-change locus, evaluation with the L^2 direct-image metric supplies its Bergman weight

$$\Phi(x) = \log \sup_{\|u\|_{L^2} \leq 1} |u(x)|^2$$

in a local relative-adjoint frame. Semipositivity follows from the dual-norm form of direct-image positivity [5, 6]; zeros of the evaluation map give allowed singularities. On the isomorphism locus of μ , the submersive differential identifies this line with $\det(T_X/\mathcal{F})$. The only extra codimension-one issue is ramification.

A prime of X vertical for the map dominates a base prime on the normal equidimensional model. Near its general point, and hence on its strict transform in W , choose coordinates

$$f = (z_1^e, z_2, \dots, z_{\dim Y}).$$

The differential contributes z_1^{e-1} . Therefore $\det(T_X/\mathcal{F})$ corresponds to the relative adjoint line twisted down by $(e-1)\{z_1=0\}$. A relative form written in total-space/base canonical frames acquires the factor $(ez_1^{e-1})^{-1}$ when restricted to a fiber. Integration on a fixed local fiber chart and the holomorphic submean estimate consequently bound every unit-norm section coefficient by $C|z_1|^{e-1}$. It follows that

$$\Phi \leq (e-1) \log |z_1|^2 + O(1).$$

After the indicated twist, the induced weight is locally bounded above at the general point of every such prime. At horizontal primes the ordinary submean estimate gives the same upper boundedness without a ramification term. These weights extend plurisubharmonically across codimension one, and then across the remaining codimension-two locus on X . They define a semipositive singular metric on $\det(T_X/\mathcal{F})$, making that line pseudoeffective. This contradicts its negative degree against α .

By movable-cone duality, the negative determinant of every cotangent tensor subsheaf is therefore pseudoeffective. The argument retains the factor $e-1$ because omitting it would construct a metric on the wrong line at ramified divisors.

6. SMOOTH COAREA METRICS AND A DOMINATED BASE

Integration along a fibration turns an anticanonical metric upstairs into an adjoint metric on its base. In this section the coefficient metric is smooth. This regularity gives a direct curvature calculation and, by coarea, an integrability exponent strictly greater than one. Together these facts produce a boundary correction supported only on base divisors whose inverse images are exceptional over the original variety.

We first establish the analytic statement, keeping curvature and integrability separate. We then construct a maximal base dominated by the anticanonical class and verify the metric hypotheses on that base. The controlled-boundary criterion converts the resulting numerical pseudoeffectivity into rational linear effectivity.

6.1. The primitive representative and horizontal strictness.

Lemma 6.1. *Let $f : \mathcal{Z} \rightarrow \mathbb{D}$ be a proper holomorphic submersion from a Kähler manifold, with fibers of complex dimension $r > 0$. Let $(\mathcal{L}, h)$ be a holomorphic line bundle with a smooth Hermitian metric of semipositive curvature. Suppose $\mathcal{E} = f_*(K_{\mathcal{Z}/\mathbb{D}} + \mathcal{L})$ is locally free and commutes with base change. Give $\mathcal{E}$ its fiberwise L^2 metric. Write D' for the $(1, 0)$ part of the Chern connection of this direct-image metric. For a local holomorphic section v with $D'v(0) = 0$, there is a smooth absolute $\mathcal{L}$ -valued $(r, 0)$ -form w representing v such that*

$$-i\partial\bar{\partial}\|v\|^2|_0 \geq f_*(c_r w \wedge \bar{w} h \wedge i\Theta_{\mathcal{L},h})|_0, \quad c_r = i^{r^2}. \quad (6.1)$$

In particular, if $i\Theta_{\mathcal{L},h} \geq cf^(i dt \wedge d\bar{t})$ on a nonempty open subset of the central fiber and its neighborhood, where $c > 0$ and $v(0)$ does not vanish identically on that subset, then the curvature pairing of $\mathcal{E}$ with $v(0)$ is strictly positive at 0.*

Proof. We use the representative construction in Berndtsson's proof of direct-image positivity [5, Proposition 4.2 and formula (4.8)]; the one-dimensional-base formulation is also [6, formula (2.1) and Lemma 2.1]. The calculation below makes clear why no strict positivity along the fibers is needed.

Write $F = f^{-1}(0)$, and fix a Kähler form ω on $\mathcal{Z}$. A smooth splitting of the relative cotangent sequence gives an absolute lift w of the relative form v . Since $dt \wedge w$ is holomorphic, write

$$\bar{\partial}w = dt \wedge \eta, \quad \partial^h w = dt \wedge \mu,$$

where ∂^h is the $(1, 0)$ part of the Chern connection. On F , η has type $(r-1, 1)$ and μ has type $(r, 0)$. Differentiating the fiberwise pairings of v with local holomorphic sections of $\mathcal{E}$ identifies the projection of μ onto the holomorphic $\mathcal{L}|_F$ -valued top forms with $D'v(0)$. These sections exhaust the top forms on F by base change. Thus $\mu|_F$ is orthogonal to the holomorphic top forms. The terms obtained by differentiating the conjugate relative section have a $d\bar{t}$ factor and do not contribute to this $(1, 0)$ derivative.

Furthermore, $(\omega \wedge \eta)|_F$ is $\bar{\partial}$ -exact. Indeed $w \wedge \omega$, of type $(r+1, 1)$ on the total space, is divisible by dt ; applying $\bar{\partial}$ and using $d\omega = 0$ gives the assertion on F . We can therefore change the lift to $w + dt \wedge \beta$, where β has type $(r-1, 0)$, so that

$$\mu|_F = 0, \quad (\omega \wedge \eta)|_F = 0 \quad (6.2)$$

for the changed lift. To verify simultaneous solvability, put $\chi = \omega \wedge \beta$. Wedge multiplication by ω is an isomorphism from $(r-1, 0)$ -forms to $(r, 1)$ -forms on F , and the Kähler identity gives

$$\bar{\partial}^* \chi = i\partial^h \beta.$$

The required equations are $\bar{\partial}^* \chi = i\mu$ and $\bar{\partial} \chi = \omega \wedge \eta$. The first right-hand side lies in the image of $\bar{\partial}^*$ because of the preceding orthogonality; the second is $\bar{\partial}$ -exact. Let $G_{\bar{\partial}}$ denote the Green operator of the coefficient $\bar{\partial}$ -Laplacian on the compact fiber. With $q = \omega \wedge \eta$, the two forms

$$\chi_1 = \bar{\partial}G_{\bar{\partial}}(i\mu), \quad \chi_2 = \bar{\partial}^*G_{\bar{\partial}}q$$

satisfy $\bar{\partial}^* \chi_1 = i\mu$, $\bar{\partial} \chi_1 = 0$, $\bar{\partial} \chi_2 = q$, and $\bar{\partial}^* \chi_2 = 0$. Indeed $i\mu$ is orthogonal to the harmonic $(r, 0)$ -forms and q is $\bar{\partial}$ -exact. Thus $\chi = \chi_1 + \chi_2$ solves both equations. Inverting $\omega \wedge$ and extending β smoothly off F proves (6.2). When $r = 1$, $q = 0$ by type and the same construction applies.

For this lift, differentiation of $\|v\|^2 = c_r f_*(w \wedge \bar{w} h)$ gives

$$\partial\|v\|^2 = c_r f_*(\partial^h w \wedge \bar{w} h).$$

Apply $i\bar{\partial}$, use $\bar{\partial}\partial^h w = \Theta_{\mathcal{L},h} \wedge w - \partial^h \bar{\partial}w$, and note that the quadratic term in $\partial^h w$ vanishes on F . To handle the remaining derivative, apply ∂ to $f_*(\bar{\partial}w \wedge \bar{w} h) = 0$; that pushforward vanishes by

type and its dt factor. At 0 the resulting identity is

$$\begin{aligned} i\bar{\partial}\partial\|v\|^2 &= f_*(c_r w \wedge \bar{w} h \wedge i\Theta_{\mathcal{L},h}) \\ &\quad - i dt \wedge d\bar{t} c_r \int_F \eta \wedge \bar{\eta} h. \end{aligned} \tag{6.3}$$

Metric pairing is denoted multiplicatively by h in these formulas. For a primitive $(r-1, 1)$ -form the last term has nonnegative sign and is a positive normalization of its squared norm. This proves (6.1).

The integrand on the right of (6.1) is nonnegative: a form of type $(r, 0)$ in complex dimension $r+1$ is decomposable, and wedging its squared form with a semipositive $(1, 1)$ -form gives a nonnegative top-degree form. On the stated patch we can replace the curvature by $cf^*(i dt \wedge d\bar{t})$ in a lower bound. Only the relative restriction of w then contributes. Its integral over an open set on which $v(0)$ is nonzero is positive. This proves strictness. $\square$

For a higher-dimensional base we apply Lemma 6.1 to small embedded disks in its tangent directions. In rank one a local holomorphic frame may be chosen Chern-normal at a specified point. Dividing (6.1) by its squared norm then gives the curvature of minus the logarithm of that norm. A lower bound by the pullback of a positive base form on one patch yields strict positivity in every nonzero tangent direction at the image point. If the relative dimension is zero and the generic rank is one, the smooth map is locally an isomorphism and this assertion follows directly from the coefficient metric.

6.2. A coarea estimate.

Lemma 6.2. *Let $f : Z \rightarrow Y$ be a surjective morphism between connected smooth projective varieties, with $\dim Z = n$ and $\dim Y = b > 0$. Let μ be a smooth nonnegative top-degree form on Z . Then $f_*\mu$ is absolutely continuous with respect to any smooth positive volume on Y , and its density belongs to $L^{1+\eta}(Y)$ for some $\eta > 0$.*

Proof. Fix Kähler metrics on Z and Y , and use their volumes and the induced fiber volumes. On the smooth locus of f let J_f be its real coarea Jacobian. In orthonormal complex frames J_f is the sum of the squared absolute values of the top complex minors of df ; equivalently, up to a fixed normalization,

$$f^*\omega_Y^b \wedge \omega_Z^{n-b} = J_f \omega_Z^n.$$

In finitely many coordinate charts it is comparable with the squared norm of the corresponding holomorphic Jacobian-minor ideal. This ideal is not identically zero, because f is dominant in characteristic zero. A log resolution of the ideal reduces small inverse powers to products of powers of coordinate absolute values. Choosing their exponents smaller than their finitely many integrability thresholds, and using compactness, gives

$$\int_Z J_f^{-\eta} dV_Z < \infty \tag{6.4}$$

for some $\eta > 0$.

Let $U \subset Y$ be a dense Zariski open over which f is smooth. Since $\mu \leq C dV_Z$, the coarea formula gives, for the density ρ ,

$$\rho(y) \leq C \int_{Z_y} J_f^{-1} dV_{Z_y} \quad (y \in U).$$

The fiber volumes are constant on each connected smooth-base component, by closedness of ω_Z^{n-b} and proper smooth local triviality. They are in particular uniformly bounded. The power-mean

inequality and coarea therefore give

$$\begin{aligned} \int_U \rho^{1+\eta} dV_Y &\leq C' \int_U \int_{Z_y} J_f^{-1-\eta} dV_{Z_y} dV_Y \\ &= C' \int_{f^{-1}(U)} J_f^{-\eta} dV_Z < \infty. \end{aligned} \tag{6.5}$$

Finally, $f^{-1}(Y \setminus U)$ is a proper analytic subset of Z , so has zero measure for the smooth form μ . There is consequently no additional measure supported on $Y \setminus U$. This proves both absolute continuity and the claimed exponent. $\square$

6.3. Extension across base divisors.

Proposition 6.3. *Let X and Z be connected smooth projective varieties, let $g : Z \rightarrow X$ be a birational morphism, and let $f : Z \rightarrow Y$ be a surjective morphism to a connected smooth projective variety of positive dimension. Put $L = -K_X$, and assume L has a smooth Hermitian metric h with semipositive curvature. Let $E = K_{Z/X}$ be the effective Jacobian divisor. Assume*

$$\text{rank } f_* \mathcal{O}_Z(E) = 1.$$

*Let $\mathcal{S}$ be the set of prime divisors G on Y such that every prime divisor of Z dominating G is g -exceptional. Suppose there is an open patch over the smooth base-change locus of f , disjoint from E , on which the curvature of g^*h dominates a positive multiple of the pullback of a Kähler form on Y . Then there is an effective integral divisor A supported on $\mathcal{S}$ and a singular metric on $M = -K_Y + A$, with psh weights ψ , such that the metric is smooth and strictly positively curved on a nonempty open subset of Y and*

$$e^{-\psi} |s_A|^2 \in L_{\text{loc}}^{1+\eta} \quad \text{for some } \eta > 0. \tag{6.6}$$

Here $|s_A|$ means the absolute value of the coefficient of the canonical section in a local holomorphic frame.

Proof. The identity section of $K_X + L$ and h determine a smooth positive volume form μ_X on X . Write ρ for the local density of $f_*(g^*\mu_X)$ in base coordinates. On a dense open $U \subset Y$, the map f is smooth, base change holds, and the canonical section s_E spans $H^0(Z_y, \mathcal{O}_{Z_y}(E))$. It gives the identification

$$f_*(K_{Z/Y} + g^*L)|_U \simeq -K_Y|_U.$$

The fiberwise L^2 norm of the corresponding coordinate anticanonical frame is ρ . This follows directly by contracting s_E with its conjugate through g^*h and integrating along the fibers; the resulting top-degree form is $g^*\mu_X$. Lemma 6.1 proves that $-\log \rho$ is psh, and the assumed patch proves strictness at its image. Smooth dependence of fiber integrals gives smoothness on U .

It remains to control the metric at the boundary. Let G be a prime component of $Y \setminus U$, and choose a prime $J \subset Z$ dominating G . Put $e_J = \text{mult}_J(f^*G) > 0$ and $h_J = \text{mult}_J E$. Near general points of J and G there are coordinates with

$$G = (t_1 = 0), \quad J = (z_1 = 0), \quad t_1 = z_1^{e_J}, \quad t_i = z_i \quad (2 \leq i \leq b).$$

Indeed $J \rightarrow G$ is generically submersive, and the unit in the first coordinate can be absorbed into z_1 . The remaining coordinates are fiber coordinates. On such a patch, the pullback volume is bounded below by a positive constant times $|z_1|^{2h_J}$ times coordinate volume. Changing variables in the first coordinate, and integrating over a fixed small fiber-coordinate polydisk, gives

$$\rho(t) \geq c |t_1|^{2((h_J+1)/e_J-1)} \tag{6.7}$$

near general points of G , on U . The denominator here is the squared Jacobian $e_J^2 |z_1|^{2(e_J-1)}$.

If $G \notin \mathcal{S}$, choose J nonexceptional for g . Then $h_J = 0$, and the exponent in (6.7) is nonpositive. Thus $-\log \rho$ is locally bounded above there. If $G \in \mathcal{S}$, choose an integer $a_G \geq \max\{0, (h_J + 1)/e_J - 1\}$. With $A = \sum_{G \in \mathcal{S}} a_G G$, the weight

$$\psi = -\log \rho + \log |s_A|^2$$

is bounded above at general boundary points. Only finitely many $G \in \mathcal{S}$ occur, since each is the image of one of the finitely many g -exceptional primes. On U the added term is pluriharmonic. The psh extension theorem therefore extends the weight across all codimension-one points under consideration. The remaining analytic subset has codimension at least two, across which psh extension is again available. One can see the needed local upper bound by using small disks through nearby points whose boundaries remain in a fixed compact subset off that analytic set, and applying the maximum principle. The frame transformations identify these extended weights as a metric of $-K_Y + A$.

The strict patch can be chosen away from A , so strict positivity is preserved there. Lastly, $g^* \mu_X$ is a smooth nonnegative volume form on the compact variety Z . Lemma 6.2 gives an exponent $1 + \eta > 1$ for its pushforward density. Since $e^{-\psi} |s_A|^2 = \rho$ almost everywhere in the compatible frames, this is exactly (6.6). $\square$

Corollary 6.4 (Smooth invariant direct image). *The conclusion of Proposition 6.3 remains valid if its ordinary rank-one hypothesis is replaced by*

$$\text{rank}(f_* \mathcal{O}_Z(K_{Z/X}))^T = 1,$$

provided that a compact torus T acts holomorphically on X and Z , acts trivially on Y , preserves f , g , and the smooth metric h , and all canonical bundles carry their natural linearizations.

Proof. On the smooth base-change locus, averaging over normalized Haar measure is a holomorphic idempotent on the adjoint direct image. Invariance of the fiber metric makes this projection orthogonal. Its image is the invariant line, and its kernel is the orthogonal holomorphic complement. Consequently the line has the quotient metric of the semipositive smooth direct image. For a section in this line, a Chern-normal holomorphic extension within the line is also Chern-normal in the full bundle, so Lemma 6.1 proves the same strictness statement. The naturally linearized Jacobian section is invariant and generates this line. Its norm is still the density ρ .

For relative dimension zero the full direct image is locally a direct sum of the coefficient lines on the sheets of an unramified cover. Its curvature is their direct sum; the same orthogonal projection argument applies, and the nonzero Jacobian section detects the strict patch. The ramification estimate, divisor correction, and coarea calculation in Proposition 6.3 do not use ordinary rank one. They therefore prove the stated extension and integrability. $\square$

6.4. From local strictness to a global strict metric. The metric just constructed is strictly positive on one open set. The next step uses the extra integrability exponent to introduce a global lower curvature bound while preserving the distinguished integrable section.

Lemma 6.5. *Let Y be smooth connected projective of positive dimension, let $A \geq 0$ be integral, and put $M = -K_Y + A$. Suppose M has psh metric weights ψ , smooth and strictly positively curved on a nonempty open set, and*

$$|s_A|^2 e^{-\psi} \in L_{\text{loc}}^{1+\eta} \quad \text{for some } \eta > 0.$$

Then M has a singular metric with a global strict curvature lower bound and with $|s_A|^2$ times its squared frame norm locally integrable. For every pseudoeffective rational divisor W , an effective rational divisor C_A supported on A makes $W + C_A$ rationally linearly equivalent to an effective rational divisor.

Proof. The curvature current of ψ has positive absolutely continuous top-degree mass, since it is smooth and positive definite on the specified open set. Boucksom's volume criterion [9] makes M big. Write $M \sim_{\mathbb{Q}} H + G$ with H ample rational and $G \geq 0$ rational. A smooth positive metric on H and the divisor metric of G give a metric on M with weights χ , curvature bounded below by a positive multiple of a Kähler form, and divisorial analytic singularities. Log resolution and compactness give a number $\tau > 0$ such that $|s_A|^2 e^{-\chi} \in L_{\text{loc}}^\tau$. Choose $\sigma \in (0, 1)$ sufficiently small that

$$\frac{1 - \sigma}{1 + \eta} + \frac{\sigma}{\tau} < 1.$$

The weights $\psi_\sigma = (1 - \sigma)\psi + \sigma\chi$ then have a global strict curvature lower bound. Moreover,

$$|s_A|^2 e^{-\psi_\sigma} = (|s_A|^2 e^{-\psi})^{1-\sigma} (|s_A|^2 e^{-\chi})^\sigma$$

is locally integrable by Hölder's inequality. Equivalently, $\mathcal{J}(\psi_\sigma) \supset \mathcal{O}_Y(-A)$. The controlled-boundary criterion, Theorem 1.1, now gives the assertion about W and C_A . $\square$

6.5. The smooth metric on the maximal dominated base. We now apply the smooth coarea construction to the maximal base of Section 5.4. That base already supplies the interpolation and the pseudoeffective divisor of vertical minima. The additional points here are a valuation proof of the model condition and a direct intersection argument for horizontal strictness. These allow us to use the original smooth metric, and hence retain the exponent $1 + \eta$ in Proposition 6.3.

Retain the hypotheses of Theorem 5.1, with effective integral divisors $N_j \sim a_j L - D$, where D is pseudoeffective and a_j is a strictly increasing unbounded sequence of positive integers. A rational map is *dominated by* L if, on a smooth resolution $X \xleftarrow{g} Z \xrightarrow{f} Y$, one has

$$g^* L - \epsilon f^* H \text{ pseudoeffective}$$

for an ample H on the projective base and a rational $\epsilon > 0$. Include the point map. This is the map formulation of the admissible systems used in Section 5.4. Indeed, an admissible system gives this inequality after resolving its base ideal. Conversely, after multiplying H so that it defines a projective embedding, its pulled-back sections push down to a subsystem of $|g_* f^* H|$; pushing down the domination inequality makes that system admissible.

The condition persists on higher source resolutions and under birational base modifications. It also persists under a generically finite change of base to which the rational map lifts dominantly: the pullback of the old ample class is big and dominates a small ample class. Finally, the joint image of two dominated maps is dominated. On a common source resolution, add suitable positive multiples of their inequalities and use the sum of the pulled-back polarizations, whose restriction to the joint image is ample. Thus the maximality construction in Section 5.4 provides a base field $\mathbb{C}(Y)$, relatively algebraically closed in $\mathbb{C}(X)$, containing the functions supplied by every dominated map.

For the resulting smooth projective diagram put

$$P = \frac{g^* N_2 - g^* N_1}{a_2 - a_1} \sim_{\mathbb{Q}} g^* L, \quad B = a_1 P - g^* N_1.$$

The interpolation argument of Section 5.4 gives $\mathbb{Q}$ -principal divisors U_j on Y with

$$g^* N_j = a_j P - B + f^* U_j, \quad U_1 = U_2 = 0. \quad (6.8)$$

For $j > 2$, the two effective divisors in $(a_j - a_1)N_2 \sim (a_j - a_2)N_1 + (a_2 - a_1)N_j$ lie in a multiple of the admissible system N_2 , so their ratio belongs to $\mathbb{C}(Y)$. In particular P has nonnegative horizontal part. If Y is a point, pushing P down already proves the theorem. Assume henceforth that $b = \dim Y > 0$.

Preparing vertical primes by their valuations. The model can be arranged so that every vertical prime not exceptional over X dominates a base prime. The flattening argument of Section 5 gives one construction. The following valuation argument gives another and explains why a base modification suffices.

On an initial model only finitely many source primes map into base codimension at least two: take an open set over which f is equidimensional and inspect the divisorial components of its complement. The valuation of such a prime restricts nontrivially to $\mathbb{C}(Y)$, since a local function vanishing on its proper center has positive order. The restricted discrete valuation is divisorial. To see this, its residue transcendence degree can drop by at most $\dim X - \dim Y$. Residues algebraically independent over the restricted residue field lift to elements algebraically independent over $\mathbb{C}(Y)$: a polynomial relation, with coefficients scaled to have minimum valuation zero, would give a residue relation. The source residue field has transcendence degree $\dim X - 1$. The restricted residue field therefore has transcendence degree at least $b - 1$, and the valuation inequality gives the opposite bound.

Choose a projective base model on which these finitely many valuations have divisorial centers, and resolve it. One can construct such a model by resolving maps to copies of $\mathbb{P}^1$ defined by $b - 1$ independent residues: the center then has dimension at least $b - 1$ and remains proper. Take a common smooth resolution of the previous source and the graph over the new base. Primes already dominating base divisors retain that property. New source primes exceptional over the old source are also exceptional over X . This proves the required preparation. The domination and field properties persist by the preceding observations. Retain P, B for their pullbacks to this model and U_j for the corresponding principal divisors on the modified base; (6.8) is unchanged.

On this prepared model define

$$w_G = \min_{J \rightarrow G} \frac{\text{mult}_J P}{\text{mult}_J f^* G}, \quad W = \sum_G w_G G. \quad (6.9)$$

The sum has finite support. Formula (6.8) makes W pseudoeffective by the same fixed-error limit as in Section 5.4: if $c_G = \min_{J \rightarrow G} \text{mult}_J B / \text{mult}_J f^* G$ and $C = \sum_G c_G G$, then $a_j W + U_j - C \geq 0$; dividing by a_j gives numerical classes converging to $[W]$.

Let $\mathcal{S}$ consist of base primes whose dominating source primes are all g -exceptional. It is finite, and the model condition makes the entire pullback of each member exceptional over X . Lemma 5.2, with its $R = P$ and $M = W$, reduces the proof to making $W + C_{\mathcal{S}}$ rationally effective for an effective correction supported on $\mathcal{S}$.

Rank one and horizontal strictness. Put $E = K_{Z/X}$. The maximality argument in Section 5.4 gives

$$\text{rank } f_* \mathcal{O}_Z(E) = 1. \quad (6.10)$$

Its concrete input is a domination inequality $b' g^* L - f^* H'$ pseudoeffective for a sufficiently ample base polarization H' . Two generic-fiber independent adjoint sections would, after an ample base twist, push down to an admissible pencil with ratio outside $\mathbb{C}(Y)$, contradicting the defining maximality. The Jacobian section supplies the nonzero rank-one generator.

Let $\alpha \geq 0$ be the smooth curvature of the original pulled-back metric on $g^* L$, normalized to represent $g^* c_1(L)$, and let r be its maximum rank. If $r = n = \dim Z$, then L is nef with positive top self-intersection and hence big, giving the required nonvanishing. Suppose $r < n$. Choose Kähler forms ω_Z, ω_Y representing ample classes. Domination gives $b'[\alpha] - [f^* \omega_Y]$ pseudoeffective for some $b' > 0$. Intersecting with $[\alpha]^r [\omega_Z]^{n-r-1}$, and using $\alpha^{r+1} = 0$, yields

$$\int_Z f^* \omega_Y \wedge \alpha^r \wedge \omega_Z^{n-r-1} = 0.$$

The integrand is nonnegative, so it vanishes pointwise. At a rank- r point, this says that $f^*\omega_Y$ vanishes on $\ker \alpha$. A nonnegative Hermitian form has zero mixed pairings against any vector on which its quadratic value vanishes. Hence

$$\ker \alpha \subseteq \ker df. \quad (6.11)$$

At a smooth point this forces $r \geq b$; rank zero is therefore excluded. The maximum-rank open set meets both the complement of E and the smooth base-change locus. On a smaller relatively compact constant-rank patch there, (6.11) gives $\alpha \geq cf^*\omega_Y$ for some $c > 0$.

Proposition 6.3 now applies with its birational map $g : Z \rightarrow X$ and fibration $f : Z \rightarrow Y$. It supplies an integral effective A supported on $\mathcal{S}$ and a semipositive metric on $-K_Y + A$, smooth and strictly positive on a nonempty open set, whose weighted density belongs to $L_{\text{loc}}^{1+\eta}$. Lemma 6.5 gives an effective C_A supported on A with $W + C_A$ rationally effective. The return lemma applies to $P \sim_{\mathbb{Q}} g^*L$: its horizontal part is nonnegative, its minimum divisor is W , and every component of f^*C_A is exceptional over X . It gives the desired section of a positive multiple of L .

This proves Theorem 5.1 using the original smooth coefficient metric. The determinant and tensor consequences remain Lemma 5.5 and Corollary 5.6. The next section retains a different bigness proof and a fiber-intersection test for horizontal positivity.

7. JET SEPARATION AND A NEF-AND-BIG DECOMPOSITION

There is another way to obtain the boundary needed for nonvanishing. A metric that is strictly positive on one open set first separates linearly growing jets at a point. This proves bigness directly by Nadel vanishing. A subsequent multiplier-ideal construction produces a nef and big divisor on a resolution, which can be perturbed by any fixed pseudoeffective class. We prove this base lemma independently, then apply it to an Iitaka fibration. The Iitaka construction also provides a different proof of the horizontal curvature bound.

7.1. A base lemma proved by jets and a nef-and-big decomposition.

Theorem 7.1. *Let Y be a smooth connected projective variety of dimension $b > 0$, let $A \geq 0$ be an integral divisor, and put $M = -K_Y + A$. Assume M has a singular Hermitian metric with psh local weights φ , smooth and strictly positively curved on a nonempty open subset, and satisfying*

$$e^{-\varphi}|s_A|^2 \in L_{\text{loc}}^{1+\gamma} \quad \text{for some } \gamma > 0. \quad (7.1)$$

For every pseudoeffective $\mathbb{Q}$ -divisor W on Y there is an effective $\mathbb{Q}$ -divisor C_A supported on $\text{Supp } A$ such that $W + C_A$ is $\mathbb{Q}$ -linearly equivalent to an effective $\mathbb{Q}$ -divisor.

Proof. We first prove that M is big, using its strict positivity only on the specified patch. Choose ample Cartier divisors H_1, A_1 so that $B_1 + A_1$ is very ample, where $B_1 = K_Y + H_1$. At a point y_0 in the patch, use coordinates z and a smooth cutoff to form a function ℓ equal to $\log \|z\|^2$ near y_0 , supported in a ball relatively compact in that patch, and zero outside it. Fix a background Kähler form ω . On the support of the cutoff error choose constants $\delta > 0$ and $C > 0$ such that $i\partial\bar{\partial}\varphi \geq \delta\omega$ and $i\partial\bar{\partial}\ell \geq -C\omega$; outside that support the added logarithmic pole has nonnegative curvature. Fix $0 < c \leq \delta/(2C)$. For large integers h and $N = \lfloor ch \rfloor$, the metric on $hM + H_1$ with weight

$$h\varphi + N\ell + \varphi_{H_1}$$

therefore has curvature bounded below by a Kähler form. Indeed the strict curvature of $h\varphi$ absorbs the cutoff error, and the positive metric on H_1 supplies a global lower bound.

Let $\mathcal{I}_h$ be this metric's multiplier ideal. Nadel vanishing [25] gives

$$H^1(Y, \mathcal{O}_Y(hM + B_1) \otimes \mathcal{I}_h) = 0.$$

Near y_0 , the original weight is smooth and

$$\mathcal{I}_{h,y_0} = \mathfrak{m}_{y_0}^{N-b+1}$$

for $N \geq b$. In fact a holomorphic monomial of total order q is integrable against $\|z\|^{-2N}$ exactly when $q > N - b$. The support of the quotient by this ideal has y_0 as an isolated component, separated from its other support by a neighborhood on which the original weight is smooth. Any jet in $\mathcal{O}_{Y,y_0}/\mathfrak{m}_{y_0}^{N-b+1}$ thus defines a global section of this quotient, taken to be zero on the other components. The vanishing lifts every such jet. The quotient has length $\binom{N}{b}$, so in fact

$$h^0(Y, hM + B_1) \geq \binom{\lfloor ch \rfloor}{b}.$$

Consequently

$$h^0(Y, hM + B_1) \geq c'h^b \quad (7.2)$$

for a constant $c' > 0$ and all sufficiently large h .

Take a fixed smooth divisor $T_0 \in |B_1 + A_1|$. We have $h^0(T_0, (hM + B_1)|_{T_0}) = O(h^{b-1})$. For completeness, choose an ample divisor H_2 such that $H_2 - M$ has a section not identically zero on any component of T_0 ; its h -th power embeds these restricted sections into those of $(hH_2 + B_1)|_{T_0}$. The latter dimensions have the asserted growth by the ample Hilbert polynomial, including the bounded zero-dimensional case when $b = 1$. The restriction exact sequence and (7.2) now imply

$$H^0(Y, hM - A_1) \neq 0$$

for large h . Thus M is big.

Write M as the sum of an ample rational class and an effective rational divisor. This gives another metric with weight ψ , curvature bounded below by a positive multiple of a Kähler form, and divisorial analytic singularities. Small inverse powers of its local divisor equation are integrable, so $e^{-\psi}|s_A|^2 \in L_{\text{loc}}^\tau$ for some $\tau > 0$. Replace φ by $(1 - \sigma)\varphi + \sigma\psi$, for a sufficiently small $\sigma > 0$. Its curvature now has a global positive lower bound. It also retains the local integrability

$$e^{-\varphi}|s_A|^2 \in L_{\text{loc}}^1. \quad (7.3)$$

Indeed the new expression is the product of the two old weighted expressions to the powers $1 - \sigma$ and σ , and Hölder applies as soon as $(1 - \sigma)/(1 + \gamma) + \sigma/\tau < 1$. We keep the notation φ for this new weight.

We now extract a nef-and-big divisor on a resolution, while retaining the integrability needed to bound every discrepancy. Fix a very ample divisor H with a positive metric. Take an integer $k > 0$ sufficiently large that $P = kM - H$ has a semipositive metric with weight $k\varphi - \varphi_H$. Put

$$B_0 = K_Y + (b + 1)H, \quad C = mP + B_0,$$

where m is a positive integer so large that $mH - B_0$ is ample. Set $\mathcal{J} = \mathcal{I}(mk\varphi)$. For $1 \leq j \leq b$, Nadel vanishing applied to $mP + (b + 1 - j)H$ gives

$$H^j(Y, \mathcal{O}_Y(C - jH) \otimes \mathcal{J}) = 0.$$

The added smooth weights leave the multiplier ideal unchanged, and the curvature is strictly positive because $b + 1 - j \geq 1$. Castelnuovo–Mumford regularity therefore makes $\mathcal{O}_Y(C) \otimes \mathcal{J}$ globally generated.

Resolve $\mathcal{J}$ and A by a smooth projective birational morphism $d : Z \rightarrow Y$, including the exceptional divisor in the resolved normal crossings support. Write $\mathcal{J}\mathcal{O}_Z = \mathcal{O}_Z(-G)$. Then $d^*C - G$ is globally generated, and

$$d^*M \sim_{\mathbb{Q}} Q + \frac{G}{mk}, \quad Q = \frac{d^*C - G + d^*(mH - B_0)}{mk}. \quad (7.4)$$

The divisor Q is nef and big: its first summand is nef, and the pullback of the ample divisor $mH - B_0$ is nef and big. This decomposition will allow us to perturb by d^*W while preserving control of the boundary coefficients.

Put

$$\Delta_0 = \frac{G}{mk} - K_{Z/Y} - d^*A. \quad (7.5)$$

We claim that Δ_0^+ is a klt boundary. At a general point of a prime $R = (z_1 = 0)$ in the resolved support, let r, a, c be its coefficients in $G, K_{Z/Y}, d^*A$, respectively. The Ohsawa–Takegoshi point-extension theorem [26] gives

$$e^{mk\varphi(d(z))} \leq C_1 |z_1|^{2r}. \quad (7.6)$$

Explicitly, point extension gives holomorphic functions with uniformly bounded $mk\varphi$ -weighted L^2 norm and $|q_{y'}(y')|^2 \geq c_1 e^{mk\varphi(y')}$ for a fixed $c_1 > 0$. Local upper bounds for the weight give ordinary sup bounds. Pullback makes every such function divisible by z_1^r ; uniform Cauchy bounds on the quotient and $y' = d(z)$ give (7.6). This argument uses the coherent multiplier ideal itself, and requires no analytic-singularity hypothesis on φ .

Changing variables in (7.3) includes the squared Jacobian of d , of order $2a$. Inequality (7.6) then forces integrability of $|z_1|^{2(a+c-r/(mk))}$. Hence

$$\frac{r}{mk} < 1 + a + c. \quad (7.7)$$

All coefficients of Δ_0 are consequently less than one. Its support has simple normal crossings, so (Z, Δ_0^+) is klt. Moreover (7.4) and $M = -K_Y + A$ imply

$$K_Z + \Delta_0 \sim_{\mathbb{Q}} -Q.$$

It remains to incorporate W without losing the strict klt inequalities. Write $Q \sim_{\mathbb{Q}} A_0 + G'$ with A_0 ample and $G' \geq 0$. For a small rational $\lambda \in (0, 1)$,

$$Q - \lambda G' \sim_{\mathbb{Q}} (1 - \lambda)Q + \lambda A_0$$

is ample. Check the pair on a common log resolution of Δ_0^+ and G' . Its crepant boundary coefficients depend affinely on λ and are strictly below one at $\lambda = 0$. Thus $(Z, \Delta_0^+ + \lambda G')$ remains klt for sufficiently small λ . This resolution is used only to check discrepancies; Q and Δ_0 continue to denote the divisors on Z . By openness of the ample cone choose a further small rational $\epsilon > 0$ so that $Q - \lambda G' + \epsilon d^*W$ is ample. Take a general effective rational representative A' of this ample class. Using a sufficiently high very ample multiple dilutes its coefficients; Bertini on the same resolution ensures that adding it preserves klt and gives it no component in common with Δ_0 . Set

$$B' = \lambda G' + A', \quad \Delta = \Delta_0 + B'.$$

Then $\Delta^+ \leq \Delta_0^+ + \lambda G' + A'$ is klt. Its positive part is big because it contains the ample rational divisor A' . Finally

$$K_Z + \Delta^+ \sim_{\mathbb{Q}} \epsilon d^*W + \Delta^-$$

is pseudoeffective. BCHM big-boundary nonvanishing [8, Theorem D] gives real-linear effectivity, and Lemma 3.1 gives rational linear effectivity. Pushing forward and dividing by ϵ proves the theorem with

$$C_A = \epsilon^{-1} d_* \Delta^-.$$

This divisor is effective and supported on A : the only negative terms introduced in (7.5) were $K_{Z/Y}$, which is exceptional, and d^*A . $\square$

7.2. The Iitaka base revisited. We apply the jet construction to the Iitaka base already used in Section 5.2. The algebraic return remains Lemma 5.2; the new point is a fiber-intersection proof that the original smooth metric is strictly positive in base directions. We also record the stronger generic-fiber rank statement behind this version of the argument.

Retain X and $L = -K_X$ with its smooth semipositive metric from Theorem 5.1. Write the given effective integral divisors as $N_m \sim mL - D$ for an unbounded set of positive integers m , with D pseudoeffective Cartier, and fix two exponents $j < q$. Use the Iitaka fibration of N_q constructed in Section 5.2. That construction settles the cases of zero-dimensional and generically finite base. In the remaining case it gives

$$X \xleftarrow{\pi} S \xrightarrow{f} Y, \quad 0 < b = \dim Y < n = \dim X, \quad u\pi^*N_q \sim f^*A_0 + I,$$

where A_0 is ample Cartier, u is a positive integer, and $I \geq 0$. The varieties are smooth projective, f has connected fibers, and every source prime mapping into base codimension at least two is π -exceptional. Every ratio from a multiple of N_q belongs to $\mathbb{C}(Y)$, which is relatively algebraically closed in $\mathbb{C}(X)$. These are precisely the model and field properties established there; no flatness of the final smooth resolution is needed.

Put $E = K_{S/X}$ and $D_S = \pi^*N_q + E$. Adding positive multiples of E does not change the sections of multiples of π^*N_q , because E is exceptional and X is normal. Thus $\kappa(D_S) = b$ and $uD_S - f^*A_0$ has an effective representative. In fact

$$\text{rank } f_*\mathcal{O}_S(\ell D_S) = 1 \quad (\ell \in \mathbb{Z}_{>0}). \quad (7.8)$$

For if this rank exceeded one, an ample base twist would yield two sections of $\ell D_S + af^*A_0$ independent over $\mathbb{C}(Y)$. Multiplication by the section of $a(uD_S - f^*A_0)$ places them in $(\ell + au)D_S$. Their ratio is transcendental over $\mathbb{C}(Y)$. Other sections of a sufficiently divisible multiple recover b algebraically independent base functions through A_0 . Products put all these ratios into one multiple of D_S , contradicting $\kappa(D_S) = b$. The canonical sections show that the rank cannot be zero. At the generic point η of Y , this gives $h^0(S_\eta, E|_{S_\eta}) = 1$, and $\pi^*N_q|_{S_\eta}$ has Iitaka dimension zero.

The interpolation and vertical-minimum construction of Section 5.2, with these indices, gives

$$R = \frac{\pi^*N_q - \pi^*N_j}{q - j} \sim_{\mathbb{Q}} \pi^*L, \quad W = \sum_T \min_{V \mapsto T} \frac{\text{ord}_V R}{\text{ord}_V f^*T} T.$$

Here R has nonnegative horizontal part and W is pseudoeffective. The rank statement above also gives the interpolation directly: for each given $m > q$, restrict the two effective divisors $(m - j)N_q \sim (m - q)N_j + (q - j)N_m$ to the generic fiber. Its one-dimensional space of sections makes their horizontal divisors equal. The horizontal slope is therefore nonnegative as m tends to infinity. The principal difference has vertical divisor, so its restriction to the projective normal generic fiber is constant; hence it is pulled back from Y . Taking vertical minima in this equality recovers the pseudoeffectivity of W by the fixed-error limit used above.

Call a base prime missed when every source prime above its generic point is π -exceptional. The model property also makes all other components of its pullback exceptional. By Lemma 5.2, it is enough to construct a weighted metric satisfying Theorem 7.1 on $-K_Y + A$, with A supported on missed primes. We now verify the only new ingredient, horizontal strictness, using intersections on the fibers.

7.3. Horizontal positivity from fiber intersections. Let $\tilde{L} = \pi^*L$ and let α be its smooth semipositive curvature form. Since D is pseudoeffective and $u\pi^*N_q - f^*A_0$ has an effective representative, $k\tilde{L} - f^*A_0$ is pseudoeffective for some $k > 0$. Let J be ample on S , put $d_0 = n - b$, and choose the largest $t \in \{0, \dots, d_0\}$ such that

$$(\tilde{L}^t J^{d_0-t} \cdot S_y) > 0$$

on a smooth general fiber. The choice exists because $t = 0$ works. These intersections are constant on the smooth base locus. Semipositivity then makes the restriction of α have rank at most t at every point of every such fiber.

The pseudoeffective comparison and nefness give

$$((k\tilde{L})^b \cdot \tilde{L}^t J^{d_0-t}) \geq ((f^* A_0)^b \cdot \tilde{L}^t J^{d_0-t}) > 0.$$

Indeed the difference of the b -th powers factors as the pseudoeffective class $k\tilde{L} - f^* A_0$ times a sum of products of nef classes. The final strict inequality is the projection formula and the definition of t . The smooth representative shows that α has rank at least $b + t$ on a nonempty open set.

A vertical null vector for a nonnegative Hermitian form is null for the full form. The vertical null space has dimension at least $d_0 - t$, so the total rank is at most $b + t$ on the smooth locus. On the preceding open set equality holds, and the full kernel is vertical. A smaller constant-rank patch, disjoint from E , therefore satisfies

$$\alpha \geq cf^* \omega_Y$$

for some $c > 0$ and a positive base form ω_Y .

The adjoint rank is one by (7.8). Proposition 6.3 now gives an integral effective A supported on missed primes and a semipositive metric on $-K_Y + A$, smooth and strictly positive on an ordinary open set, with weighted density in $L_{\text{loc}}^{1+\gamma}$ for some $\gamma > 0$. Theorem 7.1 makes $W + C_A$ rationally effective with $C_A \geq 0$ supported on A . Lemma 5.2 applied to R gives a section of a positive multiple of L . This completes the jet-based proof of Theorem 5.1.

8. A FANO-TYPE MODEL OF THE TENSOR FIBRATION

The preceding effectivity arguments correct a divisor on a smooth base. Here we instead contract the base until no correction is needed there. The additional geometric conclusion is a Fano-type model of this auxiliary base. Its contracted divisors pull back to exceptional divisors over the original variety, so a principal correction on the model still produces a section on the original variety.

We give a Fano-type proof of the tensor conversion in Corollary 5.6. Its input is a smooth connected projective X with smoothly semipositive $L = -K_X$, and nonzero sections of $(\Omega_X^1)^{\otimes p} \otimes mL$ for one fixed $p \geq 0$ and unbounded positive m . The distinctive output is the supported birational model of Theorem 8.1. We first construct the auxiliary base and verify the hypotheses of the toroidal volume theorem. We then identify exactly which base divisors its log terminal model can contract. The final passage to an anticanonical section uses the principal divisors themselves, not only their numerical classes.

8.1. A fibration tangent to the null directions. The cases $\dim X = 0$ and $p = 0$ are immediate. Suppose otherwise. Write $\mathcal{E} = (\Omega_X^1)^{\otimes p}$ and apply the determinant extraction in Lemma 5.5. Retain its determinant line M , rather than its negative: there is a nonzero map

$$M \longrightarrow \bigwedge^s \mathcal{E}$$

for some $s > 0$, and effective integral divisors

$$N_m \sim M + mL \tag{8.1}$$

for an unbounded set of positive integers m . The generic wedges used in that lemma are regular because they take values in the saturated subsheaf and its reflexive determinant. This is the determinant method of [23, Lemma 4.1] and [22, Lemma 5.1], with the unbounded exponents supplied here as part of the hypothesis.

Write $n = \dim X$, let α be the normalized curvature of L , and let ν be its maximum rank. If $\nu = n$, then L is nef and $L^n = \int_X \alpha^n > 0$, so L is big and conversion follows. Fix an ample class A and assume $\nu < n$. We will use

$$N_m \cdot [\alpha]^\nu A^{n-1-\nu} = 0. \quad (8.2)$$

Here is a differential-geometric proof of this equality. For each $\epsilon > 0$, Yau's prescribed-Ricci theorem [34] gives a Kähler metric in $[\alpha] + \epsilon A$ whose Ricci form equals the prescribed semipositive form α , up to the fixed positive normalization. Kähler symmetry identifies the mean curvature of the tangent bundle with its nonnegative Ricci endomorphism. Thus the mean curvature of every cotangent tensor power, and of its exterior powers, is nonpositive. Consequently

$$c_1(M)([\alpha] + \epsilon A)^{n-1} \leq 0.$$

Indeed, a positive degree would allow a metric on M with constant positive mean curvature, by the scalar Poisson equation. The induced mean curvature on $\bigwedge^s \mathcal{E} \otimes M^{-1}$ would then be negative definite, and the integrated Bochner formula would forbid its nonzero section. Since $\alpha^{\nu+1} = 0$, the coefficient of the lowest possible power $\epsilon^{n-1-\nu}$ is a positive constant times $c_1(M)[\alpha]^\nu A^{n-1-\nu}$ and is nonpositive. The same intersection for N_m equals it by (8.1), and is nonnegative by effectivity. This proves (8.2).

Choose a finite sum D of the N_m , allowing repetitions, whose Iitaka dimension d is maximal among such sums. These dimensions are integers in $\{0, \dots, n\}$, so a maximum exists. Choose $q > 0$ for which $|qD|$ has image dimension d . On the open set where α has rank ν , the differential of this rational map annihilates $\ker \alpha$. To see this, each member of $|qD|$ has zero intersection with $[\alpha]^\nu A^{n-1-\nu}$. A smooth local component whose tangent hyperplane failed to contain $\ker \alpha$ would retain rank ν for the restricted form, and hence contribute positive mass to that intersection. Away from the base locus, a projective hyperplane detecting a nonzero differential on $\ker \alpha$ would produce exactly such a smooth local component. Hyperplanes detect the differential of the map, so the assertion follows. In particular $d \leq \nu < n$.

Resolve the base ideal by $p_1 : X_1 \rightarrow X$, with X_1 smooth and the resolution an isomorphism off that ideal, and take the Stein factorization $f_1 : X_1 \rightarrow B$. The normal projective base B has relatively algebraically closed function field in $\mathbb{C}(X)$. There are an ample divisor A_B and an effective fixed divisor I with

$$p_1^*(qD) \sim f_1^* A_B + I. \quad (8.3)$$

Every p_1 -exceptional prime lies in $\text{Supp } I$, since its center lies in the base ideal and that ideal has positive order on the prime.

Fix $m_1 < m_2$ among the determinant exponents and put

$$G = \frac{N_{m_2} - N_{m_1}}{m_2 - m_1} \sim_{\mathbb{Q}} L, \quad J = N_{m_1} - m_1 G.$$

On any compatible higher model $\mu : W \rightarrow X$, $f : W \rightarrow Y$, with Y birational over B , there are $\mathbb{Q}$ -principal divisors T_m on Y such that

$$\mu^* N_m = m \mu^* G + \mu^* J + f^* T_m \quad (m > m_2). \quad (8.4)$$

For the principal difference of

$$(m_2 - m_1)N_m + (m - m_2)N_{m_1} \quad \text{and} \quad (m - m_1)N_{m_2}$$

is a rational function on B . Otherwise it is transcendental over $\mathbb{C}(B)$, by relative algebraic closedness, and adjoining it to the map for $|qD|$ increases the image dimension. The joint map is given by products of sections in the system for the permitted finite sum $qD + (m - m_1)N_{m_2}$, contradicting maximality. Dividing its principal divisor by $m_2 - m_1$ proves (8.4). If $d = 0$, these functions are constant: $N_m = mG + J$, so every coefficient of G is nonnegative by letting $m \rightarrow \infty$. This already proves conversion. We henceforth have $0 < d < n$.

We next establish the adjoint rank required for the volume metric. For each integer $\ell > 0$, the general fiber of f_1 satisfies

$$h^0(X_{1,b}, \mathcal{O}_{X_1}(\ell I)|_{X_{1,b}}) = 1. \quad (8.5)$$

If the dimension were larger, generic base change and a sufficiently ample base twist would give sections of $\ell I + t f_1^* A_B$ whose ratio varies on the general fiber. For $t \gg \ell$, multiply them by the section of $(t - \ell)I$. They become sections of $t p_1^*(qD)$. The same complete system recovers the base using the sections from $|t A_B|$ multiplied by s_I^t , so its image dimension exceeds d , a contradiction. The Jacobian divisor $K_{X_1/X}$ is effective and bounded by some ℓI . Its nonzero canonical section and (8.5) therefore give rank one for its full direct image. Further smooth modifications do not change this rank: their extra relative canonical divisors are effective and exceptional, and their section pushforwards equal $\mathcal{O}$ by normality. Thus on any smooth resolution $\pi : Z \rightarrow X$, $g : Z \rightarrow Y$ dominating these models,

$$\text{rank } g_* \mathcal{O}_Z(K_{Z/X}) = 1. \quad (8.6)$$

The Jacobian section identifies $g_*(K_{Z/Y} + \pi^* L)$ with $-K_Y$ on the smooth base-change locus. This is an ordinary rank assertion, involving the entire adjoint space.

8.2. The integrable correction and the model it permits. Apply birational weak toroidalization [1, Theorem 1.1], marking the nonisomorphism locus over X , and equidimensional toroidal subdivision [2, Proposition 4.4]. We obtain $\mu : W \rightarrow X$, $f : W \rightarrow Y$ with W normal projective, Y smooth projective, and f surjective, equidimensional and toroidal for strict toroidal embeddings. All modifications are birational; no reduced-fiber alteration is required. The source can be singular, and its resolution is used only for integration. The effective exceptional Weil divisor $K_{W/X}$ is supported in the toroidal boundary.

Choose a global Kähler form ω_Y on Y . By generic smoothness and cohomology-and-base-change, there is a nonempty Zariski open $U \subset Y$ on which the resolved map g is smooth and its adjoint direct image is the line identified above. The kernel calculation persists off the exceptional loci, since the map from the base to the original image is generically finite. On a small patch above U , in the maximum-rank region and away from the Jacobian divisor, it gives

$$\ker \pi^* \alpha \subset \ker dg, \quad \pi^* \alpha \geq c g^* \omega_Y \quad (c > 0).$$

The second inequality follows from the first by semipositive linear algebra and a lower bound for the nonzero eigenvalues on a smaller patch. Together with (8.6), these verify all hypotheses of the toroidal volume theorem in [29, Theorem 2.1].

Explicitly, for a base prime P and primes $S \subset W$ dominating it, put $a_S = \text{ord}_S K_{W/X}$ and $e_S = \text{ord}_S f^* P$. The theorem supplies

$$C = \sum_P \max \left\{ 0, \min_{S \rightarrow P} \left(\frac{1 + a_S}{e_S} - 1 \right) \right\} P \quad (8.7)$$

and a semipositive metric on $\Lambda = -K_Y + C$ whose local squared frame norms are integrable and whose curvature is smooth and strictly positive on a nonempty ordinary open set. Moreover C is effective rational with finite support and $f^* C$ is entirely exceptional over X . The companion theorem proves the all-valuation integrability and the extensions across divisors and codimension two; these conclusions use the precise ordinary rank and patch verified here.

In particular $\kappa(Y, C) = 0$. Indeed, a rational function with poles bounded by a positive divisible multiple of C pulls back to a function with only μ -exceptional poles. It extends across X by normality and is constant because X is connected projective. Also there is an effective big rational divisor

$$\Delta \sim_{\mathbb{Q}} \Lambda, \quad (Y, \Delta) \text{ klt}. \quad (8.8)$$

Apply Corollary 4.2 to the unweighted integrable metric on Λ . Its proof first subtracts a small ample rational class while keeping a klt representative, then adds a general representative of that

ample class. Thus the boundary here is effective, big and klt, exactly as required by the birational theorem below.

Theorem 8.1 (A model with supported contractions). *Let Y be smooth projective. Suppose $C \geq 0$ is a rational divisor with $\kappa(Y, C) = 0$, and $\Delta \geq 0$ is big and rational with (Y, Δ) klt and $K_Y + \Delta \sim_{\mathbb{Q}} C$. Then there is a birational contraction $\chi : Y \dashrightarrow Y_0$ to a projective $\mathbb{Q}$ -factorial variety of Fano type. Every prime contracted by χ lies in $\text{Supp } C$. Every pseudoeffective rational divisor on Y_0 is $\mathbb{Q}$ -linearly equivalent to an effective rational divisor.*

Proof. The big-boundary log terminal model theorem [8, Theorem 1.1] applies: the pair is klt, its boundary is effective and big, and its adjoint is pseudoeffective since it is rationally equivalent to $C \geq 0$. It gives a projective $\mathbb{Q}$ -factorial log terminal model $\chi : Y \dashrightarrow Y_0$. The map extracts no divisors. With $\Delta_0 = \chi_* \Delta$, the transformed pair is klt and $K_{Y_0} + \Delta_0$ is nef. On a common resolution $a : U \rightarrow Y$, $b : U \rightarrow Y_0$, its negativity property [8, Definitions 3.6.1 and 3.6.6] gives

$$a^*(K_Y + \Delta) = b^*(K_{Y_0} + \Delta_0) + E,$$

where $E \geq 0$ is b -exceptional and contains the strict transform of every χ -contracted prime. Choose a $\mathbb{Q}$ -principal divisor T with $K_Y + \Delta = C + T$. Using the same rational functions and coefficients downstairs, and then pulling back, cancels T on both sides. Therefore, with $C_0 = \chi_* C$,

$$a^*C = b^*C_0 + E. \tag{8.9}$$

Since Y_0 is $\mathbb{Q}$ -factorial, C_0 is rationally Cartier; it is effective, so its pullback is effective. The strict transform of each contracted prime has positive coefficient on the right of (8.9), and hence on the left. That prime must belong to $\text{Supp } C$.

The boundary Δ_0 remains big under the birational contraction. The nef adjoint is semiample by big-boundary base-point-freeness [8, Corollary 3.9.2]. The effective exceptional term E preserves the spaces of sections of all sufficiently divisible adjoint multiples, by normality. Thus

$$\kappa(Y_0, K_{Y_0} + \Delta_0) = \kappa(Y, C) = 0.$$

A semiample rational divisor with Iitaka dimension zero is rationally linearly trivial: a base-point-free multiple defines a map to a point and is therefore trivial. Hence $K_{Y_0} + \Delta_0 \sim_{\mathbb{Q}} 0$.

Write $\Delta_0 \sim_{\mathbb{Q}} A_0 + N_0$ with A_0 ample rational and $N_0 \geq 0$. For a sufficiently small rational $\lambda > 0$, the boundary $\Theta = (1 - \lambda)\Delta_0 + \lambda N_0$ is effective and klt; this follows by checking the finitely many coefficients on a common log resolution. We obtain

$$-(K_{Y_0} + \Theta) \sim_{\mathbb{Q}} \lambda A_0,$$

which is ample. This proves that Y_0 is of Fano type.

Finally, let B_0 be a pseudoeffective rational divisor on Y_0 . For a small rational $t > 0$, the class $tB_0 - (K_{Y_0} + \Theta)$ is ample. Choose a general effective rational representative U_0 of that class with sufficiently small coefficients so that $(Y_0, \Theta + U_0)$ remains klt. Its boundary is effective and big, and its adjoint is rationally equivalent to tB_0 , which is pseudoeffective. Big-boundary nonvanishing [8, Theorem D] makes this adjoint real-linearly effective. Lemma 3.1 makes it rationally linearly effective. Division by t proves the assertion for B_0 . $\square$

8.3. Returning the correction to the original variety. Apply Theorem 8.1 to (8.8). We now finish this proof of Corollary 5.6. For each base prime P set

$$g_P = \min_{S \mapsto P} \frac{\text{coeff}_S \mu^* G}{e_S}, \quad j_P = \max_{S \mapsto P} \frac{\text{coeff}_S \mu^* J}{e_S}, \quad B_g = \sum_P g_P P, \quad B_j = \sum_P j_P P.$$

These are finite rational sums. The horizontal coefficients of $\mu^* G$ are nonnegative by (8.4) and $m \rightarrow \infty$. At a component attaining the minimum g_P , that same equation and effectivity of $\mu^* N_m$

give

$$mg_P + j_P + \text{coeff}_P T_m \geq 0.$$

Thus $mB_g + B_j + T_m \geq 0$. Push this inequality by strict transform to Y_0 . Principal terms remain principal because χ extracts no divisors. Dividing numerical classes by m proves that $(B_g)_0$ is pseudoeffective. Theorem 8.1 supplies a $\mathbb{Q}$ -principal adjustment making $(B_g)_0$ effective. Use its same rational functions and coefficients on Y , and call the resulting principal divisor T_* . Then $B_g + T_*$ is effective outside the primes contracted by χ .

The divisor $\mu^*G + f^*T_*$ has nonnegative coefficient at every prime of W that is not exceptional over X . Horizontals were already treated. Above an uncontracted prime this follows from the defining minimum g_P . Above a contracted prime, every component is exceptional over X , since that prime lies in $\text{Supp } C$ and f^*C is exceptional. Equidimensionality leaves no other vertical primes. Pushing forward to X therefore gives an effective rational divisor. Its class is $\mathbb{Q}$ -linearly equivalent to $G \sim_{\mathbb{Q}} L$, because the base principal adjustment pulls back and pushes forward as a principal divisor. Clearing denominators completes the proof.

The Fano-type conclusion belongs to Y_0 , the model of an auxiliary linear-system base. The argument on X uses exactly the supported exceptional correction; it does not assert that X itself is of Fano type.

9. INVARIANT SECTIONS FROM A CURVATURE-NULL BASE

The ordinary conversion theorem does not control characters. We now construct the base from invariant ratios and retain an invariant rational section throughout the return step. The cohomological input comes from the natural anticanonical index. Here the natural linearization on $-K_X = \det T_X$ is the action induced by the differential of the action on X . The subsequent determinant, base, and metric constructions are local to the present argument.

Proposition 9.1. *Let X be a compact Kähler manifold, let a compact real torus T_c act holomorphically, and give $L = -K_X$ its natural linearization. Suppose L has a smooth semipositive metric and*

$$I_{T_c}(0) := \sum_q (-1)^q \dim H^q(X, \mathcal{O}_X)^{T_c} \neq 0.$$

There are a fixed $p \geq 0$ and unbounded positive integers m with $H^0(X, \Omega_X^p \otimes (m+1)L)^{T_c} \neq 0$.

Proof. The invariant anticanonical index theorem [28, Theorem 1.1] applies to the compact complex manifold with its natural anticanonical linearization. It gives an integer $a > 0$ and a polynomial P such that $I_{T_c}(m) = P(m)$ for every positive multiple of a , with the actual value $P(0) = I_{T_c}(0)$. In particular P is not the zero polynomial, so one degree q supplies nonzero invariant cohomology at unbounded positive divisible exponents.

Average a Kähler form and the logarithmic weights of the given metric over T_c . Hard Lefschetz with semipositive coefficients [17, Theorem 0.1] gives the equivariant surjection

$$H^0(X, \Omega_X^{n-q} \otimes (m+1)L) \longrightarrow H^q(X, K_X + (m+1)L) = H^q(X, mL).$$

The coefficient is smooth, so its multiplier ideal is trivial. Compact averaging makes the map surjective on invariant vectors. Set $p = n - q$. The shift by one is necessary because the canonical factor cancels one copy of the naturally linearized L . $\square$

Theorem 9.2. *Let X be smooth connected projective with $\chi(X, \mathcal{O}_X) \neq 0$. Let an algebraic torus T act on X , and suppose $L = -K_X$ is smoothly semipositive. For the natural anticanonical linearization, $H^0(X, mL)^T \neq 0$ for some positive integer m .*

A connected torus acts trivially on ordinary cohomology by isotopy and hence on $H^q(X, \mathcal{O}_X)$ by Hodge theory. Thus $\chi(X, \mathcal{O}_X)$ is exactly the zero value in Proposition 9.1. The point case is

immediate. For $n = \dim X > 0$, average the metric over the maximal compact torus and write its curvature as $\theta \geq 0$, normalized to represent $c_1(L)$. Let ν be its maximum rank and let ω_0 be a Kähler form. The invariant forms supplied by Proposition 9.1 give the answer immediately if their degree is zero. Otherwise determinant extraction as in Lemma 5.5 gives a linearized line A , a nonzero map $A \rightarrow \wedge^e \Omega_X^p$, and invariant sections

$$0 \neq \sigma_j \in H^0(X, A + m_j L)^T, \quad 0 < m_1 < m_2 < \cdots. \quad (9.1)$$

We will use an intersection consequence of the determinant map, rather than replace this construction by the ordinary conversion theorem.

Lemma 9.3. *If $\nu < n$, the determinant line just constructed satisfies*

$$c_1(A) \cdot [\theta]^\nu [\omega_0]^{n-1-\nu} \leq 0.$$

Proof. For each $t > 0$, Yau's prescribed-Ricci theorem [34] gives a Kähler metric in $[\omega_0 + t\theta]$ with Ricci form $2\pi\theta$. The induced mean curvature on $\wedge^e \Omega_X^p$ is negative semidefinite. If A had positive degree in this Kähler class, solving the scalar Laplace equation would give it a metric with constant positive curvature trace. The induced mean curvature on $\text{Hom}(A, \wedge^e \Omega_X^p)$ would then be negative definite, contradicting the nonzero holomorphic map by the integrated Bochner formula. Hence $c_1(A) \cdot [\omega_0 + t\theta]^{n-1} \leq 0$ for every $t > 0$. Because $\theta^{\nu+1} = 0$, the leading coefficient as $t \rightarrow \infty$ is the asserted nonpositive intersection, up to a positive binomial factor. $\square$

9.1. The invariant ratio field. Put $B_j = A + m_j L$. Form the subfield generated by ratios of invariant sections of the same line bundle $\sum_j k_j B_j$, where $k_j \geq 0$ are integers of finite support. Let F be its relative algebraic closure in $\mathbb{C}(X)$. Both fields are finitely generated: choose a transcendence basis for the smaller field; the algebraic closure of the associated purely transcendental field in the finitely generated extension $\mathbb{C}(X)$ is finite, and the intervening algebraic fields are finite as well. The field F is fixed by T . Indeed an element algebraic over the invariant subfield has finite orbit among the roots of a polynomial, and connectedness makes this orbit a point.

Finitely many ratios can be put over a common denominator by multiplication of sections. Thus one bundle $B = \sum k_j B_j$ and invariant sections of B define an image Y_0 with the smaller function field. Normalize in F , resolve the base, and obtain the invariant rational map $X \dashrightarrow Y$, with $\mathbb{C}(Y) = F$ and geometrically integral generic fiber. Let H_0 be the pullback of the image polarization; it is big and base point free. On a smooth equivariant graph resolution

$$X \xleftarrow{\pi} Z \xrightarrow{g} Y$$

there is a nonzero invariant section of $\pi^* B - g^* H_0$: divide the pulled-back defining sections by the homogeneous coordinates on the image. Hence for a chosen ample H on Y there is an integer $b > 0$ and an invariant section of $b\pi^* B - g^* H$.

If $F = \mathbb{C}$, consider

$$r_j = \frac{\sigma_j^{m_2-m_1} \sigma_1^{m_j-m_2}}{\sigma_2^{m_j-m_1}}, \quad j > 2. \quad (9.2)$$

Its numerator and denominator are invariant sections of the same multiple of B_2 , so r_j is constant. Divisor comparison and $m_j \rightarrow \infty$ show that $\text{div}(\sigma_2/\sigma_1) \geq 0$. This is the desired invariant section of $(m_2 - m_1)L$. We may therefore suppose $\dim Y > 0$.

There is a horizontal strictness patch on Z . If $\nu = n$, use any maximum-rank point away from the exceptional and critical loci. If $\nu < n$, Lemma 9.3 gives $c_1(bB) \cdot [\theta]^\nu [\omega_0]^{n-1-\nu} \leq 0$. Since $b\pi^* B - g^* H$ is effective, and all the forms involved are semipositive, it follows that

$$\int_Z g^* \omega_H \wedge \pi^* (\theta^\nu \wedge \omega_0^{n-1-\nu}) = 0.$$

At a maximum-rank point where π is an isomorphism, nonnegative linear algebra implies $\ker \pi^* \theta \subset \ker dg$. Choose such a point over the smooth base-change locus. Shrinking a constant-rank neighborhood gives

$$\pi^* \theta \geq cg^* \omega_H, \quad c > 0, \quad (9.3)$$

on a nonempty ordinary patch, disjoint from the Jacobian divisor. Rank zero is impossible here when $\dim Y > 0$.

On this equivariant model, with the trivial action on Y , the invariant part of $g_* \mathcal{O}_Z(K_{Z/X})$ has rank one. The natural Jacobian section gives rank at least one. If it were larger, global generation after a sufficiently ample base twist would give two invariant sections independent on the generic fiber. Multiplying them by a power of the invariant section of $b\pi^* B - g^* H$ puts them in $K_{Z/X} + kb\pi^* B$. Birational pushforward, using $\pi_* \mathcal{O}_Z(K_{Z/X}) = \mathcal{O}_X$, gives a ratio of invariant sections of kbB not in F , a contradiction. This is invariant rank one; no assertion about the rank of the full direct image has been made.

We have obtained the analytic data: smooth invariant coefficients, invariant adjoint rank one, the natural nonvanishing Jacobian generator, and the horizontal patch (9.3). To obtain unweighted integrability with exceptional support, we now use a separate normal toroidal model. Equivariance of that model is not required because the meromorphic data on X are already invariant.

9.2. A klt model and its unweighted volume. By weak toroidalization and equidimensional subdivision [1, 2], prepare

$$p : W \rightarrow X, \quad f : W \rightarrow Y$$

with W normal and simplicial toroidal, Y smooth, f equidimensional and toroidal, and the p -exceptional locus in the boundary. The base may be modified birationally, and the smooth equivariant graph resolution above is then replaced accordingly. For clarity, the simplicial condition means that the primitive rays of each local toric cone are linearly independent. Such toric charts are finite abelian quotients of smooth toric charts. The simplicial refinement can be made projectively using generic heights on the existing rays. Since the base cones are simplicial and source rays map to base rays or zero, each subdivided cone still maps onto a face of a base cone. Equidimensionality is preserved. Simplicial toroidal charts have quotient singularities; in particular W is $\mathbb{Q}$ -factorial and klt.

Put $D_j = \operatorname{div}(p^* \sigma_j)$ and $V = (D_2 - D_1)/(m_2 - m_1) \sim_{\mathbb{Q}} p^* L$. Equation (9.2) gives

$$D_j = D_1 + (m_j - m_1)V + \frac{f^* \operatorname{div}_Y(r_j)}{m_2 - m_1}.$$

Thus V has nonnegative horizontal part. Every vertical prime of W dominates a base prime. Its minimum divisor

$$M = \sum_Q \left(\min_{D \rightarrow Q} \frac{\operatorname{ord}_D V}{\operatorname{ord}_D f^* Q} \right) Q$$

is pseudoeffective by the coefficientwise finite-error argument following (5.2). On W one has the actual inequality $V \geq f^* M$.

The smooth metric of L defines a positive volume μ on X . On a smooth base-change open, let ρ be the coordinate density of $g_* \pi^* \mu$. This measure can equally be computed on W_{reg} ; birational changes affect only sets of measure zero. Its density is the squared norm of the Jacobian adjoint generator divided by the base canonical frame. Compact averaging on the smooth direct image is a holomorphic orthogonal projection onto the invariant line. Its norm is therefore the quotient norm. For positive-dimensional fibers, Lemma 6.1 gives

$$i\partial\bar{\partial}(-\log \rho) \geq 0,$$

with strictness on a nonempty ordinary open by (9.3). If the relative dimension is zero, connected general fibers make g birational. Over its isomorphism open the integral metric is simply the

transported coefficient metric; its curvature gives the same assertion directly, including strictness on the patch. This use of a smooth coefficient and an orthogonal quotient does not assert positivity of an arbitrary subbundle of a singular direct image.

Write $E_W = K_{W/X} \geq 0$. For a base prime Q , set

$$a_D = \text{ord}_D E_W, \quad e_D = \text{ord}_D f^*Q, \quad b_Q = \max \left\{ 0, \min_{D \mapsto Q} \frac{1 + a_D}{e_D} - 1 \right\}, \quad F_0 = \sum_Q b_Q Q.$$

This effective rational divisor has finite support, and $f^*F_0 \leq E_W$. In fact, if $b_Q > 0$, then $e_D b_Q \leq 1 + a_D - e_D \leq a_D$ for every $D \mapsto Q$. If a prime above Q is nonexceptional over X , its a_D is zero and $b_Q = 0$. Hence f^*F_0 is exceptional over X .

On $-K_Y + F_0$ consider the weight

$$\Phi = -\log \rho + \sum_Q b_Q \log |q_Q|^2,$$

initially off the indicated divisors, with q_Q a local equation of Q . At a general point of a prime D attaining the minimum, choose coordinates with $f = (z_1^{e_D}, z_2, \dots, z_{\dim Y})$. The pulled-back volume density is comparable to $|z_1|^{2a_D}$ times a smooth positive density. Integration of this chart gives

$$\rho(y) \geq c |q_Q(y)|^{2((1+a_D)/e_D-1)}.$$

Consequently Φ is locally bounded above at a general point of every omitted divisor. Plurisubharmonic extension, first across these divisors and then across codimension two, gives a semipositive metric on the rational line $-K_Y + F_0$, still smooth and strictly positive on an ordinary open.

Its squared frame norm is unweighted locally integrable. To check all valuations, take a log resolution $h : W' \rightarrow W$ of the relevant data. Integrating $e^{-\Phi}$ on the base is the same as integrating the pulled-back volume divided by the local divisor factors of f^*F_0 . The exponent divisor upstairs is

$$K_{W'} - (ph)^*K_X - h^*f^*F_0 = K_{W'/W} + h^*(E_W - f^*F_0).$$

The first summand has coefficients greater than -1 because W is klt, and the second is effective because $E_W - f^*F_0$ is effective and rationally Cartier. On a normal-crossings resolution all coefficients are therefore greater than -1 , which is exactly the local Jacobian integrability test. This includes exceptional divisors above higher-codimension centers.

Apply Corollary 5.4 to the pseudoeffective M . It gives $F_0 + \epsilon M \sim_{\mathbb{Q}} G \geq 0$ for some rational $\epsilon > 0$. Pullback and the inequality $V \geq f^*M$ make V effective after a principal correction from Y and a divisor exceptional over X . Equivalently apply Lemma 5.2 on a common resolution. The underlying rational section is a power of σ_2/σ_1 times a base function, so it is invariant. Its extension on X proves Theorem 9.2.

10. A MAXIMAL INVARIANT BASE AND SIGNED ADJOINT METRICS

There is another invariant construction when $H^q(X, \mathcal{O}_X) = 0$ for every $q > 0$. The maximal base is chosen from all invariant rational maps whose polarizations are pseudoeffectively dominated by L . Its distinguishing output is a sequence of signed boundaries, one for every positive adjoint twist. The limit of those boundaries proves the pseudoeffectivity needed for nonvanishing; the metric for a single twist would not suffice.

More precisely, if X is smooth connected projective over $\mathbb{C}$ with $H^q(X, \mathcal{O}_X) = 0$ for all $q > 0$, a compact real torus T_c acts holomorphically, and the naturally linearized $L = -K_X$ is smoothly semipositive, this construction gives $H^0(X, mL)^{T_c} \neq 0$ for some $m > 0$. Since $H^1(X, \mathcal{O}_X) = 0$, the torus preserves the isomorphism class of a very ample polarization and acts in the projective linear stabilizer of its image. A compact torus there lies in an algebraic torus after conjugation; taking its Zariski closure therefore permits equivariant projective graph resolutions. Averaging

metric weights gives a smooth invariant coefficient. The zero-index value is one. Proposition 9.1 and Lemma 5.5 supply a linearized A with $-A$ pseudoeffective and sections as in (9.1), unless a scalar invariant section has already been obtained.

10.1. Maximality and all positive adjoint ranks. Consider invariant dominant rational maps to projective bases with trivial action, subject on a resolution $\pi : Z \rightarrow X$, $g : Z \rightarrow Y$ to

$$\pi^*L - \delta g^*H \quad \text{pseudoeffective} \quad (10.1)$$

for some ample H and rational $\delta > 0$. Include the point map. This condition is unchanged by replacing the resolution. It persists under birational modification of the base or its finite Stein refinement: the pullback of an ample divisor under a generically finite map is big and dominates a small ample class. It also persists for the joint image of two such maps, by addition and a product polarization. Choose a map of maximal base dimension, take the relative algebraic closure of its field in $\mathbb{C}(X)$, and resolve the base. The resulting field is still invariant, as in the finite-orbit argument above. Every other admissible invariant ratio belongs to it: its adjoining cannot increase dimension, and the field is already relatively algebraically closed.

Fix two exponents $a < b$ in the determinant sequence, set $m = b - a$, and put $t = \sigma_b/\sigma_a$, an invariant rational section of mL . For every larger exponent i , the ratio

$$\frac{\sigma_b^{i-a}}{\sigma_i^{b-a}\sigma_a^{i-b}}$$

belongs to $\mathbb{C}(Y)$. It is a pencil in a bundle $jL + dA$ with $j, d > 0$, and $-A$ pseudoeffective makes that pencil satisfy (10.1). Thus the horizontal orders of the σ_i vary affinely in i . Their nonnegativity at unbounded i makes $D_t = \text{div}(\pi^*t)$ effective horizontally. If Y is a point, this gives the desired section directly.

On a smooth equivariant graph resolution set $E = K_{Z/X}$. For every integer $k \geq 1$ the invariant generic adjoint rank is

$$\text{rank}(g_*\mathcal{O}_Z(E + km\pi^*L))^{T_c} = 1. \quad (10.2)$$

The rational section $s_E t^k$ is regular on the generic fiber, hence supplies a nonzero invariant element. If the rank exceeded one, global generation after twisting by jg^*H would give two invariant sections in $E + km\pi^*L + jg^*H$ whose ratio is not a base function. Their effective divisors push to a pencil in $D_0 = kmL + j\pi_*g^*H$ on X ; exceptionality removes E . Pushing (10.1) forward gives an integer $b > 0$ for which $bL - D_0$ is pseudoeffective. Resolve the pushed-down pencil by $\sigma : U \rightarrow X$ and $h : U \rightarrow \mathbb{P}^1$. Its fixed part F is effective and $\sigma^*D_0 \sim h^*\mathcal{O}_{\mathbb{P}^1}(1) + F$. Consequently

$$b\sigma^*L - h^*\mathcal{O}_{\mathbb{P}^1}(1) \sim_{\mathbb{Q}} \sigma^*(bL - D_0) + F \quad \text{is pseudoeffective.}$$

This is exactly the domination premise for the new invariant pencil. Its ratio is not a base function, contradicting maximality.

We use a normal $\mathbb{Q}$ -factorial simplicial toroidal equidimensional model $p : W \rightarrow X$, $f : W \rightarrow Y$, as above, for the divisorial calculations. It need not carry the action. All positivity on invariant summands will instead be proved on the smooth equivariant model, with base change and the ranks (10.2).

10.2. The signed boundaries and their integrals. For a prime $D \mapsto Q$ on W , put

$$d_D = \text{ord}_D \text{div}(p^*t), \quad a_D = 1 + \text{ord}_D K_{W/X}, \quad e_D = \text{ord}_D f^*Q.$$

Here $a_D \geq 1$, and $a_D = 1$ for nonexceptional primes. Define

$$\gamma_Q = \min_{D \mapsto Q} \frac{d_D}{e_D}, \quad \lambda_Q(k) = \min_{D \mapsto Q} \frac{kd_D + a_D}{e_D}.$$

The first minimum records the vertical slope of t ; the second includes the Jacobian order relevant to integration. We choose the coefficient $\beta_Q(k)$ of the base correction to satisfy

$$\lambda_Q(k) - 1 \leq \beta_Q(k) < \lambda_Q(k), \quad k\gamma_Q \leq \beta_Q(k), \quad \beta_Q(k) \leq \frac{kd_D}{e_D} \quad (D \text{ nonexceptional}).$$

These inequalities have different roles. Near general points of D and Q , the local map $y_1 = z_1^{e_D}$ turns a volume density of order $2(kd_D + a_D - 1)$ into a contribution to the base density with exponent $2((kd_D + a_D)/e_D - 1)$. A prime attaining the minimum therefore gives a lower bound with exponent $2(\lambda_Q(k) - 1)$. Dividing the base density by $|y_1|^{2\beta_Q(k)}$ adds $\beta_Q(k) \log |y_1|^2$ to its negative logarithm. The first lower bound makes this corrected weight locally bounded above at Q . The divided volume upstairs has exponent $2(kd_D + a_D - 1 - e_D\beta_Q(k))$; the strict upper bound makes it integrable at each prime over Q . The toroidal calculation below will check the remaining valuations. The final upper bound ensures that the section returned to X has no pole at a nonexceptional prime. The comparison with $k\gamma_Q$ will be used after taking the limit of the correction divisors.

All four requirements are met by

$$\beta_Q(k) = \max\{k\gamma_Q, \lambda_Q(k) - 1\}, \quad B_k = \sum_Q \beta_Q(k)Q.$$

The supports lie in one finite set. Since every $a_D/e_D > 0$, both terms in the maximum are strictly below $\lambda_Q(k)$. For a nonexceptional D , both $k\gamma_Q$ and $\lambda_Q(k) - 1 \leq kd_D/e_D + 1/e_D - 1$ are at most kd_D/e_D . Finally, $\lambda_Q(k) - k\gamma_Q$ is bounded independently of k , because the minimum is over a fixed finite set. Thus

$$\beta_Q(k) < \lambda_Q(k), \quad \beta_Q(k) \leq \frac{kd_D}{e_D} \quad (D \text{ nonexceptional}), \quad \frac{B_k}{k} \longrightarrow B_\infty := \sum_Q \gamma_Q Q. \quad (10.3)$$

The chosen lower bound by $k\gamma_Q$ also gives $B_1 \geq B_\infty$. After the metrics show that $-K_Y + B_k$ is pseudoeffective for every k , the displayed limit will imply pseudoeffectivity of B_∞ and hence of B_1 .

The pair

$$(W, f^*B_k - k \operatorname{div}(p^*t) - K_{W/X}) \quad (10.4)$$

is sub-klt. Horizontal coefficients are nonpositive. A vertical coefficient equals $e_D\beta_Q(k) - kd_D - (a_D - 1) < 1$. If $e_D = 1$, both terms in the maximum defining $\beta_Q(k)$ give this coefficient at most zero. Thus the positive part is supported on the toroidal boundary. On a simplicial toroidal chart the discrepancy function has positive values $1 - b_D$ on the boundary rays, and every new ray has a positive linear combination of those values. Hence an effective boundary with coefficients below one is klt there. Subtracting an effective rational divisor only improves discrepancies. This proves (10.4), including all divisorial valuations.

Let $d\mu$ be the smooth anticanonical volume on X . On the smooth base-change locus the invariant rank-one generator of $K_{Z/Y} + (km + 1)\pi^*L$ is the relative form associated to $s_E t^k$. The squared norm in its coordinate frame is

$$\rho_k = \frac{g_*\pi^*(\|t\|^{2k}d\mu)}{dV_Y}.$$

Smooth direct-image positivity applies to the smooth invariant coefficient $(km + 1)\pi^*L$. Compact averaging is a holomorphic orthogonal projection, so the invariant line has the quotient metric. The squared frame norm on $G_k = -K_Y + B_k$ is

$$H_k = \rho_k \prod_Q |q_Q|^{-2\beta_Q(k)}.$$

Near a prime attaining the minimum in $\lambda_Q(k)$, the local map has form $y_1 = z_1^{e_D}$ and the numerator volume is comparable to $|z_1|^{2(kd_D+a_D-1)}$ times smooth volume. Therefore

$$\rho_k(y) \geq c|q_Q(y)|^{2(\lambda_Q(k)-1)}.$$

Since $\beta_Q(k) \geq \lambda_Q(k) - 1$, the weight $-\log H_k$ is locally bounded above at general points of all missing divisors. It extends plurisubharmonically there and across codimension two. Thus every G_k is pseudoeffective.

Moreover H_k is unweighted locally integrable. On a log resolution $h : W' \rightarrow W$, the divisor of poles in the integrand is

$$h^*(f^*B_k - k \operatorname{div}(p^*t)) - K_{W'/X} = h^*(f^*B_k - k \operatorname{div}(p^*t) - K_{W/X}) - K_{W'/W}.$$

Its coefficients are below one by (10.4). The normal-crossings Jacobian test and properness prove integrability on the base. This signed boundary calculation is separate from the effective exceptional correction in Section 9.2.

10.3. A global strict metric and the final section. We give the strict-metric construction with its actual sheaf and ideal conditions. From (10.1) choose a singular metric on π^*L with curvature at least $\delta g^*\omega_H$. Average its global weight difference u from the smooth invariant metric over T_c , in the sense of distributions. This preserves the quasisubharmonic lower bound. On $(m+1)\pi^*L$, add ϵu to the smooth coefficient weight. For small $\epsilon > 0$ the resulting metric h_ϵ satisfies

$$\Theta_{h_\epsilon} \geq \epsilon \delta g^*\omega_H, \quad \mathcal{J}(h_\epsilon) = \mathcal{O}_Z.$$

Small-exponent integrability on compact Z proves the ideal assertion. Consequently the singular adjoint direct-image theorem [31, Theorem 3.3.5] acts on the full ordinary direct-image sheaf. Let $U \subset Y$ be a smooth base-change open on which this sheaf is locally free. Fubini makes the fiber integral norm finite almost everywhere on U .

On U , averaging over the compact torus defines a holomorphic projection P onto the invariant summand. Invariance of h_ϵ makes P orthogonal almost everywhere, where the integral norm is finite. The dual of the quotient embeds holomorphically in the dual full bundle, and its dual norm is the restricted norm. Logarithms of these dual norms are plurisubharmonic by direct-image positivity. They therefore define the semipositive quotient metric; its rank is one by (10.2). Subtracting a base potential before applying this argument retains the lower curvature bound $\epsilon \delta \omega_H$. This proves positivity for this specified quotient, without a claim about arbitrary singular invariant subbundles.

For $k=1$ the integral norm is obtained by inserting $e^{-\epsilon u}$ in the density ρ_1 . Since u is locally bounded above, the lower density estimates remain valid up to constants. The same extension across divisors, and then codimension two, gives a metric on $G_1 = -K_Y + B_1$ with a global strict curvature lower bound. Together with the already constructed integrable semipositive metric, strong openness and Hölder's inequality give a globally strict metric on G_1 whose squared frame norms remain locally integrable. Indeed take a sufficiently small positive fraction of the strict weight; the integrable metric has some $L^{1+\eta}$ room by strong openness, and the new singular weight has small-exponent integrability.

It remains to verify the algebraic premise for the adjoint class. Every G_k/k is pseudoeffective, and $G_k/k = -K_Y/k + B_k/k \rightarrow B_\infty$. The closedness of the pseudoeffective cone gives B_∞ pseudoeffective. Since $B_1 \geq B_\infty$, B_1 is pseudoeffective as well. Choose a small ample rational A' whose smooth curvature can be subtracted from the global strict lower bound on G_1 . The resulting metric on $G_1 - A'$ remains globally strict and unweighted integrable. Apply Lemma 2.1 with $J=0$ and a sufficiently large divisible $r > 1$. On its log resolution, the fixed part divided by r , minus the relative canonical divisor, has coefficients below one. A general member of the free part, divided by r , meets this support transversely and has coefficient $1/r < 1$. It therefore gives

an effective klt $\Delta_0 \sim_{\mathbb{Q}} G_1 - A'$. Add a general effective rational representative of A' with small coefficients, transverse on a common log resolution. The resulting $\Delta \sim_{\mathbb{Q}} G_1$ is effective, big, and klt. Only the global strict bound and unweighted integrability are used here; the mixed metric need not be smooth on an open set. Now $K_Y + \Delta \sim_{\mathbb{Q}} B_1$ is pseudoeffective, so big-boundary nonvanishing [8, Theorem D] gives

$$\operatorname{div}(h) + NB_1 \geq 0$$

for some $N > 0$ and $h \in \mathbb{C}(Y)^*$, after applying Lemma 3.1. The rational section $t^N f^* h$ has no pole at horizontal primes. At a vertical prime nonexceptional over X this follows from $\beta_Q(1) \leq d_D/e_D$ in (10.3). Equidimensionality handles every such prime on W . It therefore extends to a nonzero section of NmL on X . Both t and the base function are invariant, proving invariant nonvanishing under the stated cohomology-vanishing hypothesis by this second construction.

11. COMPACT ISOMETRIC MONODROMY AND DESCENT

Let X be a smooth connected projective complex variety whose anticanonical bundle has a smooth Hermitian metric of semipositive Chern curvature. The finite-cover structure theorem supplies a compact factor to which the preceding invariant results apply. There are two legitimate orders of operation: descend invariant scalar sections, or descend twisted forms before applying an ordinary conversion theorem. We give the transport in both cases, since the second does not assume an equivariant conclusion from Theorem 5.1.

The finite-cover structure theorem [27, Theorem 2.1] supplies a connected finite étale cover $\hat{X} \rightarrow X$ whose universal cover has the holomorphic isometric splitting

$$\tilde{X} = \mathbb{C}^a \times R \times F.$$

Here R is a product of compact simply connected Ricci-flat factors, F is a product of the remaining compact projective irreducible factors, and the deck group projects to a full translation lattice on $\mathbb{C}^a$. Its action on the compact factors lies in a compact connected torus of holomorphic isometries, and the flat and Ricci-flat factors have invariant nowhere-vanishing canonical frames. Moreover $H^q(F, \mathcal{O}_F) = 0$ for $q > 0$, and $L_F = -K_F$ is smoothly semipositive. Empty products and point factors are allowed. These geometric and canonical-frame properties are supplied by the cited theorem and its proof. The separate anticanonical nonvanishing theorem of [27, Theorem 1.1] is not used here.

Remark 11.1. One can strengthen the form-vanishing part of the structure description to positive covariant tensors by a short local holonomy argument. Let S be one compact simply connected nonflat irreducible Kähler factor, with nonnegative Ricci curvature, and let $H \subset U(T_x S)$ be its holonomy group. It is connected, and its complex tangent representation is irreducible by de Rham irreducibility. If its determinant character is trivial, the canonical connection is flat and S is Ricci-flat; simple connectivity then gives a parallel canonical frame.

Otherwise the differential of the determinant character is nonzero. In the compact Lie algebra $\mathfrak{h} = \mathfrak{z} \oplus [\mathfrak{h}, \mathfrak{h}]$, trace vanishes on commutators. Some central element therefore has nonzero trace. Schur's lemma makes that element a nonzero imaginary scalar, whose exponential supplies the full scalar circle in H . On $(T_x^* S)^{\otimes p}$ this circle has weight $-p$, so it has no invariant vector for every $p > 0$. The Kähler Bochner formula with nonnegative Ricci curvature makes every holomorphic covariant tensor parallel. Hence

$$H^0(S, (\Omega_S^1)^{\otimes p}) = 0 \quad (p > 0).$$

This argument concerns all covariant tensor powers, not only exterior forms. It is independent of the descent map below, which already works for the alternating forms supplied by hard Lefschetz.

For scalar descent, apply Theorem 9.2 to F , or the compact-torus construction of Section 10. The compact action is contained in an algebraic torus: $H^1(F, \mathcal{O}_F) = 0$ preserves a very ample polarization, and a compact abelian projective linear group is conjugate into a diagonal torus. Thus a positive power of L_F has a section fixed by the projected deck action. Pull it to $\tilde{X}$ and multiply its coefficient by the matching inverse powers of the invariant canonical frames on $\mathbb{C}^a$ and R . The resulting anticanonical section is deck-invariant and nonzero, so it descends to $\hat{X}$. If F is a point, begin with its constant section. The finite étale section norm then gives a nonzero anticanonical multiple on X .

For the second order of operation, Proposition 9.1 gives a fixed p and invariant nonzero sections

$$u_m \in H^0(F, \Omega_F^p \otimes L_F^{m+1})$$

at unbounded positive m , since the actual zero index equals one. The pullback map for differential forms on the product is injective. Tensoring the coefficient with the $(m+1)$ -st inverse powers of the canonical frames on the other factors gives a nonzero form with coefficient $K_{\tilde{X}}^{-(m+1)}$. It is deck-invariant: translations preserve the flat frame, compact monodromy preserves the Ricci-flat frame, and u_m is invariant on F . It therefore descends to a nonzero section of

$$\Omega_{\hat{X}}^p \otimes (-(m+1)K_{\hat{X}}).$$

These holomorphic sections are algebraic because $\hat{X}$ is projective. If $p = 0$, we have a scalar section already. Otherwise Corollary 5.6 converts the descended forms on the actual projective cover to an ordinary anticanonical section. The invariant information was used before this ordinary conversion, not inferred from it afterward.

For clarity, the norm does not need a Galois cover. If $\nu : \hat{X} \rightarrow X$ has degree d and $s \in H^0(\hat{X}, -kK_{\hat{X}})$ is nonzero, étaleness gives $-kK_{\hat{X}} = \nu^*(-kK_X)$. Locally the product of s over the d sheets is invariant under their permutation, so it glues to a section of $-dkK_X$. At a general point every factor is nonzero, so the norm is nonzero. This is [27, Lemma 2.2], here with all its hypotheses visible.

11.1. Recovering an invariant factor section. There is also a useful converse transport after ordinary conversion. It identifies precisely the analytic role of isometric monodromy.

Proposition 11.2. *Let $d > 0$ be an integer. Suppose a compact complex manifold X' is the quotient of $\mathbb{C}^a \times W \times F$ by a group acting by product maps, whose translation projection is a full lattice in $\mathbb{C}^a$. Assume W, F are connected compact complex manifolds, the canonical bundles of $\mathbb{C}^a$ and W have invariant nowhere-vanishing frames, and monodromy on $H^0(F, -dK_F)$ preserves a Hermitian norm. A nonzero section of $-dK_{X'}$ determines a nonzero monodromy-invariant section of $-dK_F$.*

Proof. Lift the section and trivialize its coefficient on the first two factors using the invariant frames. It becomes a family of sections of $-dK_F$. This family is independent of W : its coefficients are holomorphic functions on a connected compact manifold and hence constant. More explicitly, choose finitely many evaluations at points of F , in local frames, giving an injective linear map from the finite-dimensional space $H^0(F, -dK_F)$ to a coordinate space. These evaluations also show that the family is a holomorphic map

$$v : \mathbb{C}^a \longrightarrow H^0(F, -dK_F).$$

Deck invariance and the preserved Hermitian norm make $\|v(z)\|$ periodic under the full translation lattice. It is bounded on a fundamental parallelepiped and hence everywhere. Liouville's theorem makes every coordinate of v constant. Its nonzero constant value is fixed by every projected deck transformation. The same proof includes $a = 0$. $\square$

12. EXTREMAL CHARACTERS AND INVARIANT TENSORS

The tensor conversion proved above is ordinary: its input tensors need not be invariant. This permits another way to handle compact monodromy. We first construct invariant tensors on the compact factor, then descend them before applying Corollary 5.6. The character construction below uses the extreme terms of the full equivariant Euler characteristic. It is independent of the weight-zero index calculation used in the preceding applications.

Proposition 12.1. *Let F be a smooth connected projective complex variety such that $H^0(F, \Omega_F^i) = 0$ for every $i > 0$. Suppose $L_F = -K_F$ has a smooth semipositive Hermitian metric. Let an algebraic torus T act effectively on F , and give L_F its natural tangent-determinant linearization. Then there are an integer $p \geq 0$ and unbounded positive integers k such that*

$$H^0(F, (\Omega_F^1)^{\otimes p} \otimes kL_F)^T \neq 0.$$

The trivial torus and the point variety are included.

We use the left action on sections: a group element transports a fiber value and changes its argument by the inverse action. Thus a tangent weight a induces conormal weight $-a$. This convention fixes both the orbit-degree identity and the expansions below. The use of torus characters follows the equivariant viewpoint in [24]; the present argument produces tensors to be converted on the total space, without a semiampleness assumption on the compact factor.

Proof. Assume first that T is nontrivial. Choose a general lattice cocharacter $u : \mathbb{C}^* \rightarrow T$ for which $F^u = F^T$. A T -linearized projective embedding shows that such choices exist: one avoids the finitely many hyperplanes given by differences of its weights. The fixed locus has finitely many smooth connected components P . Write $w_P \in X^*(T)$ for the character of $L_F|_P$, and put $\lambda_P = \langle w_P, u \rangle$.

The source has maximal anticanonical weight. The Białynicki-Birula decomposition [7] has a unique component P_s whose attracting set as $t \rightarrow 0$ is open. The attracting and repelling sets are affine-space bundles with ranks given by the positive and negative normal weights; see also [21, Theorem 1.5 and Corollary 7.3]. Every normal weight at P_s is positive on u . The component is proper because the action is nontrivial and effective. The natural anticanonical character is the sum of its normal tangent characters, so

$$\lambda_{P_s} > 0. \tag{12.1}$$

For every other component there is a negative normal weight. Its repelling cell supplies a nonconstant orbit with limit in that component as $t \rightarrow \infty$. Extend the *parametrized* orbit to $a : \mathbb{P}^1 \rightarrow F$ by properness. For a linearized line B the fiber weights at its two endpoints satisfy

$$\text{wt}_u(B|_{a(0)}) - \text{wt}_u(B|_{a(\infty)}) = \deg a^*B. \tag{12.2}$$

For example this follows by comparing equivariant frames at zero and infinity on $\mathbb{P}^1$. The parametrization, rather than only the normalization of its image, includes any finite stabilizer multiplicity. For an ample linearized B the difference is positive, and for the nef line L_F it is nonnegative. Iterating backward through fixed components terminates, because the ample weights strictly increase in a finite set. It can terminate only at the source. Therefore

$$\lambda_{P_s} \geq \lambda_P \quad \text{for all } P. \tag{12.3}$$

The attracting-cell limit map is a dominant rational map $F \dashrightarrow P_s$. Resolve it. Pullback of holomorphic forms along a dominant map is injective in characteristic zero, and holomorphic forms are unchanged under smooth birational modification. Consequently P_s has no positive-degree holomorphic forms. Since P_s is connected and projective, Hodge symmetry gives

$$\chi(P_s, \mathcal{O}_{P_s}) = 1. \tag{12.4}$$

Localization detects the complete source character. Apply fixed-component localization to the virtual character

$$\chi_T(F, \ell L_F) = \sum_q (-1)^q [H^q(F, \ell L_F)]$$

for positive integers ℓ ; we use the holomorphic index formula of [3, 4]. On a fixed component, write a tangent normal block as $N_a e^a$, where N_a is its underlying rank- r bundle with the weight removed. Its contribution is the factor $\lambda_{-1}(N_a^* e^{-a})^{-1}$. Expand in the direction of decreasing pairing with u . If $\langle a, u \rangle > 0$, this factor is

$$\sum_{j \geq 0} \text{Sym}^j(N_a^*) e^{-ja};$$

the leading term is 1 and all others have negative pairing. If $\langle a, u \rangle < 0$, the same factor is

$$(-1)^r \det(N_a) e^{ra} \sum_{j \geq 0} \text{Sym}^j(N_a) e^{ja};$$

even its leading term has strictly negative pairing. These identities follow by factoring the highest exterior term in the second case. Every normal character pairs nontrivially with u , so each fixed exponent receives only finitely many contributions; equivalently one works in the descending Laurent completion over the character field of the sublattice pairing to zero.

By (12.3), every nonsource contribution is strictly below $\ell \lambda_{P_s}$, including components with $\lambda_P = \lambda_{P_s}$. At the source the only term at that pairing is the constant term of each positive normal factor. Thus the entire part at the highest pairing is

$$e^{\ell w_{P_s}} \chi(P_s, \ell L_F|_{P_s}). \quad (12.5)$$

This records the full T -character, not merely its restriction to the chosen one-parameter subgroup. Riemann–Roch makes its coefficient a polynomial in ℓ whose value at zero is 1 by (12.4). The coefficient is therefore nonzero for all sufficiently large positive ℓ . Hence the character ℓw_{P_s} occurs in at least one actual cohomology group $H^q(F, \ell L_F)$ for each such ℓ .

Canceling source characters produces tensors. Only finitely many fixed components can occur as sources while u varies among general lattice directions. Zero lies in the convex hull of their characters. Otherwise strict rational separation, followed by a small general perturbation, gives a lattice direction pairing negatively with every source character. Its own source contradicts (12.1). Since the characters are integral, there are nonnegative integers b_s , not all zero, with

$$\sum_s b_s w_{P_s} = 0. \quad (12.6)$$

For each source with $b_s > 0$, choose a general direction having that source. Formula (12.5) applies for all sufficiently large ℓ in each of these finitely many choices. Passing to an infinite common subsequence fixes the cohomological degrees q_s in which the characters occur.

Average a Kähler form and the logarithmic weights of the given line metric over the maximal compact torus T_c . Smooth semipositivity is preserved. Hard Lefschetz with the smooth coefficient $(\ell + 1)L_F$ [17, Theorem 0.1] gives T_c -equivariant surjections

$$H^0(F, \Omega_F^{h-q_s} \otimes (\ell + 1)L_F) \longrightarrow H^{q_s}(F, \ell L_F), \quad h = \dim F. \quad (12.7)$$

The target identification is equivariant because $K_F + (\ell + 1)L_F = \ell L_F$ with the natural linearizations. Projecting a lift to its character space by weighted Haar averaging therefore lifts the exact character ℓw_{P_s} .

Tensor the lifted nonzero sections with multiplicities b_s and use the alternating inclusions $\Omega_F^j \hookrightarrow (\Omega_F^1)^{\otimes j}$ in characteristic zero. Their tensor product is nonzero at the generic point, since F

is integral. By (12.6) its character is zero. Its tensor degree and line exponent are

$$p = \sum_s b_s(h - q_s), \quad k = (\ell + 1) \sum_s b_s.$$

The first is fixed and the second tends to infinity. Invariance under T_c is equivalent to T -invariance in the finite-dimensional algebraic section representation. This proves the nontrivial-torus case.

If T is trivial, Hodge symmetry gives $\chi(F, \mathcal{O}_F) = 1$. Riemann–Roch for ℓL_F is a polynomial with this constant term, so some cohomology group is nonzero for all sufficiently large ℓ . Fix its degree along an infinite subsequence and apply (12.7). For a point take $p = 0$ and the constant section for every $k > 0$. $\square$

12.1. Descent before ordinary conversion. We finish this method by specifying the tensor descent. Use the finite-cover structure statement as in Section 11: on the universal cover of a connected finite étale cover $\hat{X} \rightarrow X$ there is an isometric holomorphic product

$$\tilde{X} = \mathbb{C}^a \times R \times F.$$

Here R is the compact Ricci-flat product, F is projective with no positive-degree holomorphic forms, and $-K_F$ is smoothly semipositive. The chosen deck subgroup acts by translations on the flat factor, fixes the canonical frame on R , and acts on F through a connected compact torus of holomorphic isometries. Point factors are allowed. These are the geometric structure and frame assertions, independent of the anticanonical nonvanishing conclusion.

For completeness, the compact action has the algebraic realization needed by Proposition 12.1. Since $H^1(F, \mathcal{O}_F) = 0$, it preserves a very ample line up to isomorphism and hence acts projectively in its complete linear-system embedding. The representation is continuous, as one sees by tracking a projective frame of points on the embedded variety. A connected compact abelian subgroup of the projective linear group is conjugate into a diagonal torus. Its Zariski closure is an algebraic torus preserving F ; quotient by its ineffective kernel if necessary. A diagonal projective torus has a linear lift after fixing one diagonal coordinate, so an ample linearization is available. The action on $-K_F$ remains the intrinsic tangent-determinant action.

Pull each invariant tensor u_k from the proposition to $\tilde{X}$ and use the cotangent inclusion from the product projection to F . If ω_{flat} and ω_R are the invariant canonical frames of the other factors, then

$$u_k \otimes (\omega_{\text{flat}} \otimes \omega_R)^{-k} \in H^0(\tilde{X}, (\Omega_{\tilde{X}}^1)^{\otimes p} \otimes (-kK_{\tilde{X}}))$$

is nonzero and deck-invariant. It descends to $\hat{X}$ and is algebraic by projectivity. Corollary 5.6 now gives a nonzero anticanonical plurisection on this actual projective finite cover. Finally the finite étale norm recalled in Section 11 gives a nonzero positive anticanonical multiple on X . Thus the only invariant objects needed for this method are the tensors before descent; the conversion and its Fano-type auxiliary model are ordinary.

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